

Virasoro Constraints under Projectivization

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Abstract

We prove that full ordinary descendant Virasoro constraints pass from a smooth projective complex base to the projectivization of any algebraic vector bundle of rank at least two. The bundle need not split and satisfies no positivity requirement. Assuming the full constraints on the base, the conclusion includes every genus, each individual integral curve class, and all cohomology insertions, including primitive and odd classes. The result also applies successively to towers of projective bundles.

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1 Introduction

Virasoro constraints organize the descendant Gromov–Witten invariants of a smooth projective variety into differential equations for a single generating function. A basic structural question is whether these equations pass from a base to the total space of a geometric fibration. We prove that they pass through projectivization of an arbitrary algebraic vector bundle.

For a smooth connected projective complex variety Y , let Z_Y be its total ordinary descendant potential: the exponential of the connected stable-map potentials, with genus weight $\hbar^{g-1}$, ordinary cotangent-line classes at the markings, and monomials Q^d indexed by the actual effective classes $d \in H_2(Y; \mathbb{Z})$. We use all of $H^*(Y; \mathbb{C})$ with its cohomological parity and Poincaré pairing. The Virasoro operators are the normally ordered operators associated to the first Hodge grading

$$\mu_Y|_{H^{p,q}(Y)} = (p - \tfrac{1}{2} \dim_{\mathbb{C}} Y) \text{id}, \quad R_Y = c_1(TY) \cup .$$

Their zero-mode constant is $C_Y = \chi(Y)/16 - \text{str}(\mu_Y^2)/4$, and the other modes have no scalar correction. Section 2 specifies the loop-space and quantization conventions completely. Write $V(Y)$ for the coefficientwise identities

$$L_k^Y Z_Y = 0 \quad (k \geq -1).$$

Thus $V(Y)$ includes every genus, individual integral curve class, and finite list of descendants, including primitive and odd insertions.

Theorem 1.1. *Let B be a smooth connected projective variety over $\mathbb{C}$, and let E be an algebraic vector bundle of rank $r \geq 2$ on B . Let $X = \mathbb{P}_B(E)$ parametrize one-dimensional subspaces of the fibers of E . Then*

$$V(B) \implies V(X).$$

The bundle is arbitrary: it need not split or admit a filtration by line bundles, and it is subject to no positivity condition. The base is allowed to have nonsemisimple quantum cohomology and arbitrary Hodge types. In particular the assertion applies successively to towers of projective bundles once the constraints hold on the initial base. The conclusion concerns the ordinary theory of each total space.

1.1 Historical context and related results

The Virasoro conjecture in Gromov–Witten theory extends the differential constraints for intersection numbers on moduli spaces of stable curves arising in Witten’s conjecture and Kontsevich’s theorem [21, 15]. Eguchi–Hori–Xiong proposed the corresponding constraints for quantum cohomology [4]. The extension to general Hodge types and odd cohomology, including Katz’s correction of the grading, is described by Eguchi–Jinzenji–Xiong [5]; see also Getzler’s account [7]. Liu–Tian prove the constraints in genus zero [18]. In all genera, Givental’s quantization formalism and Teleman’s reconstruction theorem establish them for targets with generically semisimple quantum cohomology [9, 10, 20]. Okounkov–Pandharipande prove them for smooth target curves, including odd insertions [19]. Together with Theorem 1.1, this gives the full constraints for every projective bundle over a smooth projective curve, and for towers of such bundles.

The closest transfer theorem is due to Coates–Givental–Tseng [3]. They prove that the Virasoro constraints hold for the base of a toric bundle if and only if they hold for its total space, for toric bundles constructed from sums of line bundles. Their proof combines ancestor localization with Brown’s mirror theorem [1]; split projective bundles are among its examples. For arbitrary vector bundles, Fan [6] proves all-genus reconstruction and Chern-class dependence of projective-bundle invariants. His argument uses the projective completion $\mathbb{P}_B(E \oplus \mathcal{O})$, which is also the auxiliary space used below. Reconstruction of invariants by itself does not establish compatibility with the Virasoro differential operators.

Iritani–Koto [14] construct a mirror theorem and a decomposition of the quantum D -module for arbitrary projective bundles. Koto [17] proves a mirror theorem for nonsplit toric bundles. These results concern genus zero. In particular, Iritani–Koto’s equivalence of generic semisimplicity for a projective bundle and its base, combined with Givental–Teleman, already gives the conclusion of Theorem 1.1 when the base has generically semisimple quantum cohomology. The transfer proved here also covers bases whose quantum cohomology is not semisimple.

As in Fan’s reconstruction, we use a projective completion of the bundle as an auxiliary space. The localization calculation adapts the ancestor organization of Coates–Givental–Tseng, including their use of the genus-zero cone and ancestor identities. We give the additional gluing argument required when the orbit lines are parametrized by a positive-dimensional variety. The mechanism for passing constraints between the two fixed components uses Iritani’s equivariant shift operator [13]. A local logarithm of that shift cancels equivariant differentiation in the grading. The resulting spectral modes can be quantized and their equations continued between the components. Quantum Riemann–Roch [2] identifies the local equations with the ordinary constraints. The argument uses virtual localization [12] and the ancestor–descendant formalism [16, 8, 9]; the necessary forms of these results are recalled at their points of use.

1.2 The proof mechanism

Twisting E by a sufficiently negative line bundle leaves X unchanged and makes E^* globally generated. Set

$$W = \mathbb{P}_B(E \oplus \mathcal{O}).$$

Let $\mathbb{C}^*$ scale the trivial summand, and let λ be its equivariant parameter. Its fixed components are B and X . Write y for the variable measuring tautological degree and retain separate variables Q^β for the full integral base classes. The proof has five stages.

1. *Express the auxiliary ancestors using the two fixed theories.* Localization factors the ancestor

potential of W through the product of the inverse-Euler-twisted ancestor potentials of B and X . The transformation is the quantization of an upper symplectic series $R(z)$. The same R factors the genus-zero fundamental solution. Section 4 proves both statements together, keeping the families of orbit lines as evaluation correspondences. Localization provides a relation involving both fixed theories; a second structure is needed to transfer constraints from one to the other.

2. *Separate the shift operator into spectral blocks.* The genus-zero shift operator translates λ by the loop variable z . Its coefficients, at each base degree, are polynomial in y . After reducing modulo z and positive base degree, its distinct spectral values are $\lambda + h$, where

$$h^r(h + \lambda) = y.$$

There are $r + 1$ branches at a generic value of y . Near $y = 0$, one tends to the eigenvalue zero and corresponds to B ; the other r tend to λ and together correspond to X . Section 5 establishes this description without diagonalizing the quantum cohomology of B .

3. *Construct quantizable modes on the blocks.* The equivariant grading contains $\lambda\partial_\lambda$. On each block the logarithm of the shift contains $z\partial_\lambda$. Subtracting λ/z times this logarithm removes coefficient differentiation. The remaining grading defines Virasoro-type loop operators. Section 6 constructs their projections and logarithms coefficientwise and compares them at $y = 0$ with the fixed-component operators. Section 7 uses quantum Riemann–Roch to identify the latter, up to a triangular change of modes, with the ordinary modes on B or X .
4. *Continue equations between branches.* For fixed genus, base class, and insertions, the connected ancestor invariants of W are polynomial in y . Moreover, ancestor powers are bounded independently of fiber degree. Every coefficient of a quantized equation therefore involves finitely many coefficients of the spectral modes and is an actual identity of germs. The covering $h \mapsto h^r(h + \lambda)$ is connected off its branch values. Its monodromy transports the equations from the B branch to every branch. Adding the r branches of the X cluster and returning to $y = 0$ gives the ordinary equations for X up to scalar.
5. *Fix the scalar normalization.* Quantization is projective, so its covariance has only determined the equations up to insertion-independent scalars. The commutators $[L_{-1}, L_1] = -2L_0$ and $[L_0, L_k] = -kL_k$ remove those scalars and give exactly the prescribed constant in L_0 . Section 8 proves this calculation and completes the transfer.

Two features of the argument are useful beyond the localization formula itself. Subtracting the block logarithm of a difference operator can turn a grading that differentiates parameters into a loop operator over the coefficient ring. Ancestor bounds then let identities for such operators continue coefficientwise, even when continuation of an entire quantized transformation has not been defined. In the present setting these two steps supply the passage from the ordinary constraints on one fixed component to those on the other.

2 Descendant potentials and quantization

We fix the ordinary theory and its operator normalization before introducing the auxiliary equivariant target. Quantized changes of frame are naturally defined up to a scalar. We therefore record both the exact Virasoro constraints and the weaker, projective form that can be transported between frames. The scalar ambiguity will be removed at the end of the proof.

2.1 Cohomology, curve classes, and descendants

Let Y be a smooth connected projective complex variety of dimension d . Its state space is the full super vector space $H_Y = H^*(Y; \mathbb{C})$, with parity given by cohomological degree modulo two and pairing

$$\eta_Y(a, b) = (a, b)_Y = \int_Y a \cup b.$$

Every tensor product, permutation, contraction, and derivative below uses the Koszul convention. In particular, the coevaluation tensor is the categorical inverse of this pairing, without an additional parity involution. Define even endomorphisms

$$\mu_Y|_{H^{a,b}(Y)} = (a - d/2) \text{id}, \quad R_Y = c_1(TY) \cup. \quad (2.1)$$

Thus μ_Y uses the first Hodge index. Poincaré duality and the Hodge type of $c_1(TY)$ give $\mu_Y^* = -\mu_Y$, $R_Y^* = R_Y$, and $[\mu_Y, R_Y] = R_Y$.

Choose a homogeneous Hodge basis (ϕ_a) with $\phi_0 = 1_Y$. For $j \geq 0$, let t_j^a have the parity of ϕ_a , and put

$$t_j = \sum_a t_j^a \phi_a, \quad t(z) = \sum_{j \geq 0} t_j z^j.$$

These are formal supercommuting variables; the aggregate insertion t_j is even. The connected, ordinary, unreduced descendant potentials are

$$\begin{aligned} F_g^Y(t) &= \sum_{\substack{\beta \text{ effective} \\ m \geq 0}} \frac{Q^\beta}{m!} \int_{[\overline{\mathcal{M}}_{g,m}(Y, \beta)]^{\text{vir}}} \prod_{i=1}^m \left(\sum_{j \geq 0} \psi_i^j \text{ev}_i^* t_j \right), \\ Z_Y(t) &= \exp \left(\sum_{g \geq 0} \hbar^{g-1} F_g^Y(t) \right). \end{aligned} \quad (2.2)$$

Only stable maps contribute. Here ψ_i is the cotangent class at the marked point of the stable map. It is not an ancestor class. The exponential includes disconnected domains and the empty domain.

The Novikov symbols retain the actual effective integral classes $\beta \in H_2(Y; \mathbb{Z})$, including $\beta = 0$; multiplication is $Q^{\beta_1} Q^{\beta_2} = Q^{\beta_1 + \beta_2}$. Fixing an ample integral class gives the Novikov completion: only finitely many labels occur below each fixed ample degree. All identities are read coefficientwise in this completion, at finite order in the descendant variables, and with the coefficientwise Laurent interpretation in $\hbar$. In particular, classes are not identified merely because their divisor degrees agree. The finiteness of effective homology labels of bounded degree follows, for example, from the finite-type Chow spaces of cycles of bounded degree in a projective embedding.

2.2 The ordinary Virasoro operators

On the loop space and its standard polarization use

$$\begin{aligned} \mathcal{H}_Y &= H_Y((z^{-1})), \quad \Omega_Y(f, g) = \text{Res}_{z=0}(f(-z), g(z))_Y dz, \\ \mathcal{H}_Y &= H_Y[z] \oplus z^{-1} H_Y[[z^{-1}]]. \end{aligned} \quad (2.3)$$

The infinitesimal symplectic operators are

$$\ell_{-1,Y} = z^{-1}, \quad \ell_{0,Y} = z \partial_z + \frac{1}{2} + \mu_Y + \frac{R_Y}{z}, \quad \ell_{k,Y} = \ell_{0,Y}(z \ell_{0,Y})^k \quad (k \geq 1). \quad (2.4)$$

Their infinitesimal symplectic property follows from the adjoint identities after (2.1), including the sign of z in the residue pairing.

For an even infinitesimal symplectic operator A on a paired loop space with pairing η , write

$$\mathcal{Q}_A(f) = \frac{1}{2}\Omega(f, Af).$$

Use homogeneous Darboux coordinates for the displayed polarization, with positive coordinates q_j^a and dual negative coordinates $p_{j,a}$. Normal ordering quantizes a pp monomial as $\hbar$ times the corresponding second derivative, a pq monomial as the first-order operator with multiplication before differentiation, and a qq monomial as multiplication by that monomial divided by $\hbar$. Derivatives are left derivatives and all reorderings have their graded signs. We denote the resulting operator, after the dilaton translation

$$q(z) = t(z) - z1_Y, \quad (2.5)$$

by $\text{op}_\eta(A)$. The negative coordinates $p_{j,a}$ in this paragraph are unrelated to the tautological divisor p introduced below.

With this sign convention the string operator is

$$L_{-1}^Y = \text{op}_{\eta_Y}(\ell_{-1,Y}) = -\frac{\partial}{\partial t_0^0} + \sum_{j \geq 0, a} t_{j+1}^a \frac{\partial}{\partial t_j^a} + \frac{(t_0, t_0)_Y}{2\hbar}. \quad (2.6)$$

For all other modes set

$$\begin{aligned} L_k^Y &= \text{op}_{\eta_Y}(\ell_{k,Y}) \quad (k \neq 0), \\ L_0^Y &= \text{op}_{\eta_Y}(\ell_{0,Y}) + C_Y, \quad C_Y = \frac{\chi(Y)}{16} - \frac{1}{4} \text{str}(\mu_Y^2). \end{aligned} \quad (2.7)$$

The supertrace uses even minus odd cohomological parity. There are no scalar corrections in positive modes. We write $\mathbb{V}(Y)$ for the full set of identities

$$L_k^Y Z_Y = 0 \quad (k \geq -1). \quad (2.8)$$

These equations use every genus, every effective integral class, and all insertions in H_Y , including odd and primitive classes.

2.3 Projective constraints and changes of frame

Definition 2.1. An even infinitesimal symplectic operator A is *satisfied projectively* by a potential Z if

$$Z^{-1} \text{op}_\eta(A) Z$$

is independent of the descendant or ancestor variables. In this expression $\text{op}_\eta(A)$ acts on Z . Such a scalar may depend on the Novikov variables, equivariant parameters, and $\hbar$.

This formulation discards exactly the scalar ambiguity of quadratic quantization. It is useful for comparing different paired state spaces. For a symplectic map M between them, denote its quantized action, when defined in the completions used here, by U_M . Our group-action convention is the one in the following covariance formula. With the Hamiltonian sign above, its infinitesimal form uses $-\text{op}_\eta(A)$ for the action of $\exp(A)$.

Lemma 2.2 (Projective covariance). *Suppose M and M^{-1} admit quantized actions, and A and MAM^{-1} admit coefficientwise quadratic quantizations. If η and η' are the source and target pairings, then*

$$U_M \text{op}_\eta(A) U_M^{-1} = \text{op}_{\eta'}(MAM^{-1}) + \text{a scalar.} \quad (2.9)$$

Consequently A is satisfied projectively by Z if and only if MAM^{-1} is satisfied projectively by $U_M Z$. Multiplying either potential by an invertible scalar does not change this condition.

Proof. The commutator of two normally ordered quadratic operators is the quantization of the corresponding quadratic Hamiltonian bracket, together with the scalar arising from the double contractions. In super coordinates that scalar is the corresponding supertrace. Exponentiating this identity gives (2.9); equivalently one can use the projective quantization formalism of [9]. The same argument applies between paired spaces after a linear change of Darboux coordinates. The last assertions follow because these operators do not differentiate coefficient parameters or $\hbar$. $\square$

2.4 Coefficientwise finiteness near zero curve variables

Some later mode matrices have infinitely many positive powers of z , and can also contain finite powers of $z\partial_z$. We interpret their residue identities entrywise in the polarized mode coordinates. The arrays need not act on every uncompleted loop-space input; only the coefficientwise actions specified here are used. The following elementary rule specifies the formal setting for their local quantizations; the continuation argument will establish its separate finiteness assertion at generic fiber parameter.

Lemma 2.3. *Suppose a potential has finite descendant-index support at fixed curve coefficient, $\hbar$ power, and variable order, and has $\hbar$ powers bounded below at fixed curve coefficient and variable order. Let A be an even infinitesimal symplectic mode operator, linear over the coefficient parameters and independent of $\hbar$. Assume its powers of z have a finite lower bound and its order in $z\partial_z$ is finite at each filtration coefficient. Then $\text{op}_\eta(A)$ acts coefficientwise on this potential. The same holds with additional formal filtration variables. Quantized actions of transformations equal to the identity modulo positive filtration, whose logarithms satisfy these mode bounds, are defined by their formal exponential series in this setting.*

Proof. For the pp part, only finitely many differentiated indices have nonzero coefficients in the potential. For the pq part, a fixed differentiated index bounds the possible multiplied index because the mode powers have a lower bound. The qq part has only finitely many pairs of indices at each filtration coefficient. Euler differentiation in z multiplies mode coefficients by polynomials in their indices and does not change these bounds. At fixed positive filtration degree only finitely many terms of an exponential occur. These observations also make the cross-polarization contractions in the covariance formula finite coefficientwise. $\square$

Ordinary descendant potentials have this support property: for fixed genus, class, and number of marks, sufficiently high powers of the cotangent classes vanish on the finite-dimensional stable-map space. It also holds for the twisted descendants used below, where the torus acts trivially on the target and the cotangent classes remain ordinary classes. Ancestor potentials have the stronger bound supplied by the dimension of the moduli space of stable curves. Stability ensures the stated Laurent property after exponentiation: degree-zero genus-zero components consume marked points, whereas positive-degree components consume positive Novikov degree.

The standard splitting and cotangent-class identities used in the paper are tensor identities with the diagonal given by the categorical coevaluation. Their marked-point proofs therefore apply to full cohomology with precisely these signs. Characteristic-class multiplications are even; quantum Riemann–Roch and the two-mark shift identities likewise use arbitrary evaluation classes and this diagonal. Throughout, a splitting adds the actual integral curve classes. These conventions permit the standard formulas to be used without discarding odd insertions or merging Novikov labels.

3 The master space and its fixed theories

We place the base and its projectivization inside a single smooth projective variety with a torus action. Its fixed components will carry inverse-Euler twists of their ordinary theories. This section specifies their curve labels and pairings, and recalls the ancestor and fundamental solution conventions needed for the localization comparison.

3.1 Geometry and integral curve labels

Tensoring E by a line bundle does not change the projective bundle of one-dimensional subspaces. Choose such a twist so that E^* is globally generated, and continue to denote the resulting bundle by E . Set

$$X = \mathbb{P}_B(E), \quad W = \mathbb{P}_B(E \oplus \mathcal{O}_B).$$

Let $T = \mathbb{C}^*$ act with weight zero on E and weight one on $\mathcal{O}_B$, and let $\lambda \in H_T^2(\text{pt})$ be the class of the weight-one representation. The fixed locus is $W^T = X \sqcup B$, where the second component is the section corresponding to the summand $\mathcal{O}_B$. Write π for either projective-bundle projection and put $p = c_1^T(\mathcal{O}_W(1))$. The ordinary normal bundles and their torus weights are

F	N_F	$n_F = \text{rk } N_F$	w_F	$W_F = n_F w_F$	j_F
X	$\mathcal{O}_X(1)$	1	1	1	0
B	E	r	-1	$-r$	1

(3.1)

The restrictions of the divisor are

$$p|_X = c_1(\mathcal{O}_X(1)), \quad p|_B = -\lambda.$$

Here w_F is the common torus weight on the normal fibers; the integers j_F are recorded for the fixed-section curve factors in the shift operator below. The normal-bundle descriptions follow from the relative tangent space $\text{Hom}(L, V/L)$ at a line $L \subset V$.

Lemma 3.1. *For $P = X$ or W , the map*

$$H_2(P; \mathbb{Z}) \longrightarrow H_2(B; \mathbb{Z}) \oplus \mathbb{Z}, \quad d \longmapsto \left(\pi_* d, \int_d c_1(\mathcal{O}_P(1)) \right) \quad (3.2)$$

is an isomorphism, including torsion. An effective class has an effective base pushforward β and a nonnegative tautological degree l .

Proof. The structure group of a projective bundle acts trivially on the integral homology of its fiber. In total degree two the homology Serre spectral sequence therefore has only the base term $H_2(B; \mathbb{Z})$ and the fiber term $H_2(\mathbb{P}^{s-1}; \mathbb{Z}) = \mathbb{Z}$, where $s \geq 2$ is the bundle rank. The integral class $c_1(\mathcal{O}_P(1))$ pairs to one with the fiber line. Consequently the fiber generator survives: its image

in total homology cannot be zero or a nontrivial multiple quotient, as either possibility would contradict that pairing. The edge filtration gives an exact sequence

$$0 \longrightarrow \mathbb{Z}[\text{fiber line}] \longrightarrow H_2(P; \mathbb{Z}) \xrightarrow{\pi_*} H_2(B; \mathbb{Z}) \longrightarrow 0.$$

Pairing with $c_1(\mathcal{O}_P(1))$ is a retraction on its first term; together with π_* it gives (3.2). This integral argument does not discard torsion. Finally, the dual bundles E^* and $E^* \oplus \mathcal{O}_B$ are globally generated, so the corresponding tautological line bundles are globally generated. Their degrees on effective curves are nonnegative, and proper pushforward preserves effective curve classes. $\square$

We write $Q^\beta y^l$ for the actual class corresponding to (β, l) . Under the inclusion $X \hookrightarrow W$ these two coordinates are unchanged, and under $B \hookrightarrow W$ a class β becomes $(\beta, 0)$. Thus the fixed theories and the master-space theory have compatible full integral labels. We may allow all pairs of an effective base class and an integer $l \geq 0$ as formal labels, assigning coefficient zero when a pair is not effective for the target in question.

Choose an ample integral class H on B . For a sufficiently large integer M , the class $c_1(\mathcal{O}_P(1)) + M\pi^*H$ is ample for both projective bundles. We complete in the degree

$$l + M \int_\beta H.$$

There are finitely many effective labels of bounded degree, as in Section 2. Unless specified otherwise, positive Novikov filtration means positive degree for this completion, so it includes positive fiber degree even when the base class is zero. The separate completion by positive base degree will be used only when the fiber parameter y is treated as a variable on a complex domain.

3.2 Equivariant pairings and inverse-Euler twists

The equivariant projective-bundle formula makes $H_T^*(W)$ free over $\mathbb{C}[\lambda]$, with basis

$$\pi^* \phi_a p^j, \quad 0 \leq j \leq r,$$

for a homogeneous Hodge basis (ϕ_a) of H_B . Its equivariant integration pairing is perfect over $\mathbb{C}[\lambda]$. Indeed, integration over the projective fibers gives an antitriangular matrix in the powers of p , whose antidiagonal blocks are the ordinary Poincaré pairing of B .

After extending scalars to $\mathbb{C}(\lambda)$, restriction to the fixed locus is an isomorphism of paired spaces

$$\begin{aligned} (H_T^*(W), \eta_W) \otimes_{\mathbb{C}[\lambda]} \mathbb{C}(\lambda) &\cong \bigoplus_{F=B, X} (H_F \otimes \mathbb{C}(\lambda), \eta_F^t), \\ \eta_F^t(a, b) &= \int_F \frac{a \cup b}{e_T(N_F)}. \end{aligned} \tag{3.3}$$

The unit restricts to the sum of the two fixed units. Every normal weight is nonzero, so the Euler classes in this formula are invertible over $\mathbb{C}(\lambda)$.

The corresponding twisted Gromov–Witten theory on F inserts

$$e_T(R^\bullet \rho_* f^* N_F)^{-1} \tag{3.4}$$

in each stable-map integral. Here ρ and f are the universal curve and universal map, and the inverse Euler class is extended multiplicatively to the indicated K -theory class. The torus acts trivially

on F and with weight w_F on N_F . We use subscripts F, tw for its potentials and fundamental solutions. Its degree-zero three-point pairing is exactly η_F^t .

Give p and λ Hodge bidegree $(1, 1)$. In a homogeneous polynomial basis the first-Hodge grading acts on basis vectors, while differentiation in λ records the degree of equivariant coefficients when needed. The virtual dimension identities can be expressed in this first degree: algebraicity forces the sum of the Hodge-degree differences of insertions in a nonzero invariant to be zero. This remains true equivariantly, by finite-dimensional approximations or by fixed-locus integration. It is distinct from the ordinary cohomological-degree count used to bound fiber degree later.

3.3 Ancestors and the fundamental solution

We give the same definitions for an ordinary theory, the equivariant theory of W , or one of the twisted fixed theories. Denote its pairing by η , its Novikov monomial for a class d by $\mathbf{q}^d$, and its stable-map integrals by brackets. Let u be a formal even primary background. For distinguished insertions $a_1, \dots, a_m$, set

$$\langle a_1, \dots, a_m \rangle_{g,u} = \sum_{\substack{d \text{ effective} \\ n \geq 0}} \frac{\mathbf{q}^d}{n!} \langle a_1, \dots, a_m, \underbrace{u, \dots, u}_n \rangle_{g,m+n,d},$$

where d runs over effective classes and only stable-map terms are included. Descendant insertions in these brackets use the map cotangent classes.

For $2g - 2 + m > 0$, there is a stabilization map

$$\text{st}_{g,m} : \overline{\mathcal{M}}_{g,m+n}(Y, d) \longrightarrow \overline{\mathcal{M}}_{g,m}$$

which forgets the map and the n background marks and stabilizes the remaining curve. Define $\bar{\psi}_i = \text{st}_{g,m}^* \psi_i$ for $1 \leq i \leq m$. The ancestor potential at background u is

$$\begin{aligned} \bar{F}_g(u; t) &= \sum_{\substack{d \text{ effective}, m, n \geq 0 \\ 2g-2+m > 0}} \frac{\mathbf{q}^d}{m!n!} \left\langle \prod_{i=1}^m \left(\sum_{j \geq 0} \bar{\psi}_i^j \text{ev}_i^* t_j \right) \prod_{a=m+1}^{m+n} \text{ev}_a^* u \right\rangle_{g,m+n,d}, \\ \mathcal{A}(u; t) &= \exp \left(\sum_{g \geq 0} \hbar^{g-1} \bar{F}_g(u; t) \right). \end{aligned} \tag{3.5}$$

The bracket in this display denotes integration of the displayed product, with the twist when appropriate. All terms with curve-unstable distinguished data are omitted, even when the stable-map space itself exists. The genus-one term with no distinguished marks is therefore omitted as well. Since the ancestors come from $\overline{\mathcal{M}}_{g,m}$, their total power exceeds the dimension $3g - 3 + m$ only when the corresponding product vanishes.

Definition 3.2. The fundamental solution $S(u, z) = \text{id} + O(z^{-1})$ is defined by

$$\eta(b, S(u, z)a) = \eta(b, a) + \left\langle b, \frac{a}{z - \psi} \right\rangle_{0,u}, \quad \frac{1}{z - \psi} = \sum_{j \geq 0} \psi^j z^{-j-1}. \tag{3.6}$$

The pairing term supplies the unstable two-point contribution.

The genus-zero splitting and cotangent-class identities give

$$S(u, -z)^* S(u, z) = \text{id}, \quad z \partial_v S(u, z) = (v \star_u) S(u, z), \quad (3.7)$$

where $\star_u$ is the quantum product and ∂_v differentiates the primary background. For clarity, the genus-zero cone $\mathcal{L}$ is the formal Lagrangian graph of dF_0 in the Darboux coordinates (q, p) , with $q = t - z1$. Its point with descendant input ϵ has positive coordinate $\epsilon - z1$, and the tangent space there is the graph of the Hessian of F_0 at ϵ . The genus-zero string, dilaton, and topological recursion identities say that each tangent space T is tangent along $zT \subset \mathcal{L}$. At primary coordinate u that tangent space is $S(u, z)^{-1} \mathcal{H}_+$; see [11]. The ancestor–descendant correspondence in our group-action convention is

$$\mathcal{A}(u) = c(u) U_{S(u)} Z, \quad (3.8)$$

where $c(u)$ is independent of ancestor variables. These formulas follow from the splitting identity for $\psi_i - \psi_i$ and its genus-zero specializations; see [16, 8, 9]. They hold with the super contractions of Section 2 and with the inverse-Euler twists above.

We use (3.8) only on the fixed theories, with u of positive total Novikov filtration. There $S(u, z) - \text{id}$ has positive filtration, and every Novikov coefficient has only finitely many negative powers of z : only finitely many background marks occur at that coefficient, and the map cotangent classes on the fixed targets are nilpotent. Thus the quantized action is covered by Lemma 2.3. For the master space we abbreviate

$$S_W = S_W(0, z), \quad \mathcal{A}_W = \mathcal{A}_W(0).$$

Its rational loop-variable expansions and the quantized action used to compare it with the fixed theories will be established directly by localization in the next section.

4 Ancestor localization for the master space

The torus action on $W = \mathbb{P}_B(E \oplus \mathcal{O})$ separates its Gromov–Witten theory into contributions from B and X . We need this separation at the level of ancestor potentials, together with the corresponding factorization of the genus-zero fundamental solution. We abbreviate the fixed paired space of (3.3) by

$$H^{\text{fix}} = H_B \oplus H_X, \quad \eta^{\text{fix}} = \eta_B^t \oplus \eta_X^t.$$

Restriction identifies this paired space with $H_T^*(W)$ after inverting λ . All series below use the full curve labels $Q^\beta y^l$.

Proposition 4.1 (Ancestor localization). *There are classes $u_F \in H_F$ with coefficients of positive Novikov filtration, for $F = B, X$, and a symplectic power series $R(z) \in \text{End}(H^{\text{fix}})[[z]]$, with $R - \text{id}$ of positive Novikov filtration, such that, writing*

$$S_f(z) = S_{F, \text{tw}}(u_F, z), \quad S_{\text{bl}}(z) = \bigoplus_{F=B, X} S_f(z),$$

one has

$$S_W(z) = R(z) S_{\text{bl}}(z), \quad (4.1)$$

$$\mathcal{A}_W(0) = c U_R \prod_{F=B, X} \mathcal{A}_{F, \text{tw}}(u_F), \quad (4.2)$$

where c is an invertible scalar independent of ancestor variables. For every curve coefficient, $S_W(z)$ is the expansion at infinity of a rational function of z . In (4.1) that rational function is expanded at $z = 0$; the equality is in Laurent series with a finite lower bound at each curve coefficient. Both R and its inverse admit quantized actions in the coefficientwise completion used here.

The master space and its fixed-graph geometry also occur in Fan's reconstruction of projective-bundle invariants [6, §§3–4]. For the ancestor factorization we adapt the localization argument of Coates–Givental–Tseng [3, §3]. Their toric-bundle proof has a finite set of orbit directions in each fiber. Here the orbit lines form the projective bundle X , so their contributions are cohomology correspondences. We first establish the geometry and gluing formula for these families. After this step, the genus-zero identities determine R , and stabilization of the localization graphs gives its quantized action. This supplies, for the action at hand, the leg-moduli and virtual-class details left open in [3, Remark 3.7].

4.1 The moving components and their gluing

Call an irreducible component of a torus-fixed stable map a *moving leg* if its image is not contained in $W^T = B \sqcup X$. A connected part mapping into a fixed component will instead be kept as a stable-map factor for that component, including its internal boundary strata.

Lemma 4.2. *A moving leg has degree $m \geq 1$ on a line joining a point $x \in X$ to its image $b = \pi(x) \in B$. Its map from $\mathbb{P}^1$ is totally ramified over x and b , and its markings and nodes lie at these two points. For any specified marking and attachment status at the endpoints, the leg moduli is a smooth proper Deligne–Mumford stack, the μ_m -gerbe $q_m : \mathfrak{L}_m \rightarrow X$ of m th roots of $\mathcal{O}_X(-1)$. Its endpoint evaluation maps are q_m and $\pi \circ q_m : \mathfrak{L}_m \rightarrow B$.*

At an endpoint in F , the tangent line of the leg has equivariant first Chern class

$$a = \frac{w_F \lambda}{m} + \nu,$$

where ν is an ordinary degree-two class on the leg stack. Every node involving a moving leg has nonzero smoothing character.

Proof. After a finite cover of the torus, its action lifts to the domain of a fixed stable map. A nontrivial torus action on a complete irreducible curve has rational normalization. Since the torus acts trivially on B , the projection of a moving component is constant. In a projective fiber $\mathbb{P}(E_b \oplus \mathbb{C})$, the closure of every nonfixed orbit is the line joining $[0 : 1]$ to a unique point $[v : 0] \in \mathbb{P}(E_b)$. An equivariant finite map to this orbit closure is, in coordinates at the endpoints, $t \mapsto t^m$. A self-node would identify its two fixed points, which have distinct images. Thus the moving component itself is smooth, and the assertions about its special points follow.

Varying the endpoint $[v]$ gives X as the parameter space of orbit lines. On the gerbe $\mathfrak{L}_m$, let M be the universal line with $M^{\otimes m} \cong q_m^* \mathcal{O}_X(-1)$. The universal cover is

$$\mathbb{P}(M \oplus \mathcal{O}) \longrightarrow \mathbb{P}(q_m^* \mathcal{O}_X(-1) \oplus \mathcal{O}) \longrightarrow W, \quad [s : t] \longmapsto [s^m : t^m].$$

Locally every leg has this form, and its automorphisms are exactly the deck group μ_m . This identifies its moduli with the stated root gerbe, which is smooth and proper over the projective variety X . Its ordinary class also agrees with its fixed virtual class: on a line L in a fiber,

$$T_{W/B}|_L \cong \mathcal{O}(2) \oplus \mathcal{O}(1)^{\oplus(r-1)}, \quad f^* \pi^* T_B \cong T_{B,b} \otimes \mathcal{O}_{\mathbb{P}^1}.$$

The tangent sequence therefore gives $H^1(\mathbb{P}^1, f^*T_W) = 0$. The ambient pointed-map space is smooth at the leg, and taking the fixed part of its deformation space gives the tangent space of this smooth gerbe.

The universal cover gives, on $\mathfrak{L}_m$, the tangent classes

$$a_X = \frac{\lambda + q_m^* p|_X}{m}, \quad a_B = -a_X.$$

Their scalar characters are $+\lambda/m$ at X and $-\lambda/m$ at B . A node joining a leg to a fixed-map factor has this nonzero character. At a node joining two legs, both branches approach the same fixed component, so the smoothing character is $w_F \lambda(1/m + 1/m')$, again nonzero. $\square$

The geometry is illustrated in Figure 1. Nonsplitting of E affects the global family of lines and its cohomology classes, but does not change the description of an individual leg or the nonzero characters in the lemma.

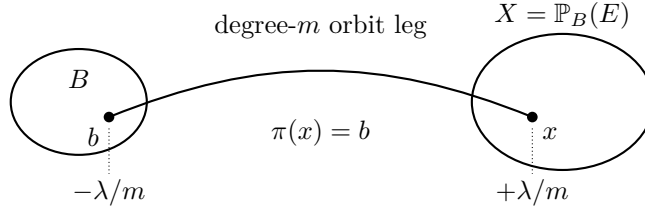


Figure 1: The two endpoints of a moving leg. The displayed characters are the scalar parts of its tangent classes. Choosing $x \in X$ determines the orbit line; the multiplicity- m covers retain the finite stabilizer μ_m .

We now apply virtual localization [12]. To make its use with these families explicit, first distinguish all branches at gluing nodes and divide by graph automorphisms afterward. A torus-fixed deformation cannot smooth any of the nodes cut in Lemma 4.2: the smoothing parameter has nonzero character. The fixed loci are therefore obtained by matching leg endpoints and stable maps to B or X along their evaluation maps. This description holds in families. Indeed, degree zero over B forces the projection of a rational leg family to factor through its base, and the local monomial description then applies over a trivialization of E . On a stable-map piece with image in F , the lifted torus acts trivially on the domain: its automorphism group as a pointed stable map is finite. Fixed deformations of that piece consequently remain maps into F .

For completeness, the compatibility of obstruction theories can be read directly from normalization. Denote the normalized pieces by C_α and the cut nodes by q . The map deformation complexes, relative to their prestable domains, fit into the triangle

$$R\Gamma(C, f^*T_W) \longrightarrow \bigoplus_{\alpha} R\Gamma(C_\alpha, f_\alpha^*T_W) \longrightarrow \bigoplus_q T_{W, f(q)}.$$

Cut branches are marked on the pieces. Passing to absolute map deformations adds their pointed-domain deformation and automorphism complexes; releasing a cut node adds its smoothing line $T_{q,+} \otimes T_{q,-}$. The ambient virtual tangent complex, restricted to this fixed gluing locus, consequently has class

$$[\mathbb{T}_\Gamma^{\text{vir}}] = \sum_{\alpha} [\mathbb{T}_\alpha^{\text{vir}}] - \sum_q [\text{ev}_q^* T_W] + \sum_q [T_{q,+} \otimes T_{q,-}]. \quad (4.3)$$

Here each piece carries its pointed stable-map obstruction theory; for a leg this is the ambient theory restricted to $\mathfrak{L}_m$. The identity comes from the displayed compatible triangle, so its fixed part gives the obstruction theory of diagonal matching, not only an equality of virtual ranks. On fixed-map pieces the fixed part is their theory in F ; on legs it is $T_{\mathfrak{L}_m}$. The matching terms are T_F , and every smoothing line is moving. The fixed virtual class is therefore the diagonal virtual pullback of these classes, including when two legs meet or a graph has a cycle. The moving part contains $R\pi_* f^* N_F$ on each fixed-map piece, the moving leg complexes, the matching terms $-N_F$, and the smoothing lines. These are the factors required by virtual localization.

In particular, cutting a leg off a fixed-map vertex contributes the normal matching numerator $e_T(N_F)$ and the smoothing denominator

$$\frac{1}{a - \psi}. \quad (4.4)$$

Here ψ belongs to the fixed-map vertex and a to the leg. The matching numerator is exactly what makes the contraction use the inverse of η_F^t . All remaining leg data are cohomology correspondences independent of the descendants on the vertex. There is also an endpoint version of this rule: if an external marked point replaces the attachment to a fixed-map vertex, there is no smoothing denominator, its cotangent class is $-a$, and contraction with η_F^t cancels the normal matching numerator. This convention includes endpoints with no fixed-map component.

The ordinary class in a need not descend along $X \rightarrow B$. We therefore expand it on the leg stack before any pushforward. If $a = a_0 + \nu$, with $a_0 = w_F \lambda / m \neq 0$, then

$$\frac{1}{a - s} = \sum_{j \geq 0} \frac{(-\nu)^j}{(a_0 - s)^{j+1}}. \quad (4.5)$$

The sum is finite. More explicitly, let C be a correspondence coefficient on a cut piece, with endpoint map e to F , and put $h_j = e_*(\nu^j C)$, with its virtual and Euler factors included. The label h will retain this entire finite family of moments, together with the scalar a_0 . For any function K at this flag define

$$K(a)h := \sum_{j \geq 0} \frac{1}{j!} \partial_\alpha^j K(\alpha) \Big|_{\alpha=a_0} h_j. \quad (4.6)$$

The symbol α is a dummy scalar: the derivative acts on K alone, holding C and all h_j fixed, even if they depend on λ . For example the later expression $S_f(a)h/(a+z)$ means (4.6) with $K(\alpha) = S_f(\alpha)/(\alpha+z)$. At several attachment flags use the joint moments and independent dummy scalars. A scalar tangent class and an ordinary cohomology vector are the special case in which only h_0 is present. This convention makes the correspondence formulas precise even when the tangent class does not descend to F .

All graph sums in this section are coefficientwise finite. Each moving leg has positive tautological degree, so its multiplicity and the number of legs are bounded at fixed ample degree. There are finitely many splittings into effective full curve labels, and stability bounds the remaining degree-zero pieces once genus and the distinguished markings are fixed. The fixed-map moduli spaces bound the powers of their cotangent classes. These observations also justify the finite nilpotent expansions and all termwise pushforwards used below.

4.2 The genus-zero factor

We first construct R from two-point invariants. Fix $F = B$ or X . An *end* at F is a genus-zero unmarked tree attached to a fixed-map vertex by a moving leg, with the vertex itself omitted.

Let $\epsilon_F(s)$ be the sum of its contributions, including the denominator $(a-s)^{-1}$ at the attachment. It is a cohomology-valued power series in s , of positive Novikov filtration. Likewise a *tail* has one external primary insertion $b \in H_T^*(W)$; write its contribution as $h_b/(a-s)$, with summation over tails understood. The moment data h_b include the curve monomial, the stack and symmetry factors, and the linear dependence on b . These definitions include an external insertion directly at a leg endpoint.

End denominators have infinite Taylor expansions. Accordingly, in identities involving these backgrounds we use the completion $\hat{\mathcal{H}}_+ = H_F[[z]]$ of the positive space, with localized Novikov coefficients, and $[\cdot]_+$ denotes the nonnegative part. The usual cone identities extend to these backgrounds coefficientwise: positive filtration bounds their number of insertions, and nilpotence of the map cotangent classes bounds the indices which can contribute.

In the twisted theory of F , let u_F be the primary parameter of the tangent space to the genus-zero cone at descendant background ϵ_F . Recall the particular genus-zero facts that we use. The string, dilaton, and topological recursion relations make the cone overruled, and its tangent spaces are $S_{F,\text{tw}}(u, z)^{-1}\hat{\mathcal{H}}_+$ in this completion [11]. The parameter u_F is equivalently determined by

$$x_F(z) := [S_f(z)(\epsilon_F(z) - u_F)]_+ \in z\hat{\mathcal{H}}_+, \quad S_f(z) = S_{F,\text{tw}}(u_F, z). \quad (4.7)$$

Indeed, applying S_f to the cone point makes it belong to $z\hat{\mathcal{H}}_+$, and $[S_f(z)(-z + u_F)]_+ = -z$. The vanishing of the constant term in (4.7) has linearization $-\text{id}$ with respect to u_F at zero Novikov degree. It therefore determines u_F uniquely by the formal implicit equation, and u_F has positive filtration.

The tangent space is the graph of the Hessian of the genus-zero potential. Thus its two-point function depends on ϵ_F only through u_F . In the following formula the bracket sums the stable genus-zero terms with any number of additional $\epsilon_F(\psi)$ insertions:

$$\frac{\eta_F^t(b, c)}{w + z} + \left\langle \frac{b}{w - \psi}, \frac{c}{z - \psi} \right\rangle_{0, \epsilon_F}^{F, \text{tw}} = \frac{\eta_F^t(S_f(w)b, S_f(z)c)}{w + z}. \quad (4.8)$$

At primary background this is the standard two-point fundamental solution identity; the Hessian description gives it at the present background. It follows alternatively by genus-zero topological recursion. The metric term records the unstable two-point convention. Every occurrence of this identity is rational in w, z at a fixed curve coefficient.

For c supported on F , localize the matrix element $\eta(b, S_W(z)c)$. If the input lies on a fixed-map vertex, all unmarked outgoing trees provide its background ϵ_F . The insertion b is either on that same vertex, as $b|_F$, or in a tail with attachment $h_b/(a-\psi)$. If the input itself is a marked leg endpoint, the endpoint rule supplies the metric term in (4.8), with denominator $a+z$. Taking also the coefficient at w^{-1} in that identity to treat a primary slot gives

$$\eta(b, S_W(z)c) = \eta_F^t(b|_F, S_f(z)c) + \sum_{\text{tails}} \frac{\eta_F^t(S_f(a)h_b, S_f(z)c)}{a + z}. \quad (4.9)$$

This proves (4.1): explicitly,

$$(R(z)^*b)_F = b|_F + \sum_{\text{tails}} \frac{S_f(a)h_b}{a + z}, \quad (4.10)$$

expanded at $z = 0$. The denominators are invertible there, so R is upper triangular. The fixed-theory S_f is finite in negative powers of z at each curve coefficient, and the tails give rational

functions. This also proves the claimed rationality of S_W . Since S_W and S_{bl} are symplectic, their rational identities imply

$$R(-z)^* R(z) = \text{id}. \quad (4.11)$$

Every nonidentity term in (4.10) has a moving leg, so $R - \text{id}$ has positive Novikov filtration.

Two further genus-zero identities identify the translation and the edge contraction for the quantized action. They will allow us to recognize the higher-genus localization formula without computing the individual leg Euler factors.

First use the one-point function $J_W(-z) = -z S_W(z)^{-1} 1$. The string equation identifies it with

$$J_W(-z) = -z 1 + \sum_{\beta, l} Q^\beta y^l (\text{ev}_1)_* \left(\frac{[\overline{\mathcal{M}}_{0,1}(W, (\beta, l))]^{\text{vir}}}{-z - \psi_1} \right),$$

where only stable one-marked terms are included. In the expansion at zero, its regular part on F is $-z + \epsilon_F(z)$. Indeed, a marked leg endpoint supplies an end, whereas a marking on a fixed-map vertex contributes only strictly negative powers of z . Hence

$$\epsilon_F(z) = z + [J_W(-z)|_F]_+.$$

Apply S_f and take the regular part. Because S_f has only nonpositive powers, inserting or omitting an inner regular-part projection does not alter the final regular part. The string equation gives $[S_f(z)z]_+ = z + u_F$; using the factorization already proved yields

$$\bigoplus_F x_F(z) = z(1 - R(z)^{-1} 1). \quad (4.12)$$

Next a *bridge* is an unmarked genus-zero tree with two attachments to fixed-map vertices, omitted from the tree, and moving legs at both attachments; a single leg is allowed. Its contribution has denominators $(a - s)^{-1}(a' - s')^{-1}$. Apply $S_f(a)$ and $S_{f'}(a')$ to its two correspondence slots, and denote the sum of these dressed contributions by $D(s, s')$. A tensor is here regarded as an operator by the twisted pairing; the first argument belongs to the output slot.

Localize the two-point function of W , including its metric term. If no moving leg separates the marked points, the contribution is the block identity (4.8). Otherwise the path between them determines a unique bridge, and (4.8) at its two ends includes the cases in which either head is an unstable marked endpoint. Consequently

$$\frac{S_W(w)^* S_W(z)}{w + z} = S_{\text{bl}}(w)^* \left(\frac{\text{id}}{w + z} + D(-w, -z) \right) S_{\text{bl}}(z). \quad (4.13)$$

Together with (4.1), this gives

$$D(-w, -z) = \frac{R(w)^* R(z) - \text{id}}{w + z}. \quad (4.14)$$

The numerator vanishes at $w = -z$ by (4.11). Thus the right side is a power series at $(w, z) = (0, 0)$, as is the dressed bridge expression. We may first establish these equalities with independent rational variables and then take these Taylor expansions.

4.3 From descendants at the flags to ancestors

We have constructed the upper factor and identified its translation and bridge kernel. To obtain (4.2), it remains to express every surviving vertex of a localization graph by its twisted ancestor theory. We record the two ancestor identities needed for that conversion, including their dependence on the background.

For a fixed target with twisted pairing η^t , let $\mathcal{A}(\epsilon; t)$ denote the mixed potential with distinguished insertions $t(\bar{\psi})$ and additional insertions $\epsilon(\psi)$. Here the ancestors forget the additional marks. The parameter ϵ is of positive filtration, and u and $x(z) = [S(u, z)(\epsilon(z) - u)]_+$ are specified by (4.7).

Lemma 4.3 (Vertex conversion). *In the summed vertex correlators with background ϵ , an insertion at a distinguished flag transforms as*

$$\frac{h}{a - \psi} \mapsto \frac{S(u, a)h}{a - \bar{\psi}}. \quad (4.15)$$

The transformation remains valid with additional powers of the ancestor class at that flag and with other distinguished flags; all evaluation classes at the flag are included in h before applying $S(u, a)$. Moreover, up to a scalar independent of t ,

$$\mathcal{A}(\epsilon; t) = \mathcal{A}(u; t + x). \quad (4.16)$$

Proof. These are the ancestor identities of Getzler and Kontsevich–Manin [8, 16], in the form used in [3, §3.5, Proposition 3.6]. We give the arguments because the background here has descendants.

The identity

$$\frac{1}{a - \psi} = \frac{1}{a - \bar{\psi}} + \frac{\psi - \bar{\psi}}{(a - \psi)(a - \bar{\psi})}$$

reduces the first assertion to the splitting divisor for $\psi - \bar{\psi}$. This divisor separates a genus-zero twig carrying the indicated mark, any subset of the background marks, and its attaching node. Splitting the virtual class contracts its two-point function with the rest of the vertex. Summing all such twigs, and adding the direct term, gives $S(u, a)h$ by the primary-slot instance of (4.8). The ancestor classes at the original distinguished marks are pulled back from the stabilized curve, so they do not enter this twig calculation. This proves (4.15), successively at every distinguished flag. For tangent classes with nilpotent part, apply the scalar identity and then the finite derivatives specified in (4.5).

For the second assertion write $\epsilon = u + (\epsilon - u)$ and expand multilinearly, distinguishing the two kinds of extra marks. First forget only the primary u marks. The same splitting-divisor argument converts the remaining descendant insertions to $x(\bar{\psi}) = [S(u, z)(\epsilon(z) - u)]_+|_{z=\bar{\psi}}$. At this intermediate stage, the original ancestors still forget the x marks, whereas the new ancestors retain them.

The two choices give equal correlators. A boundary divisor in their cotangent-class difference has a rational tail carrying one original mark and $k \geq 1$ of the marks to be forgotten. Its stable-curve factor is $\overline{\mathcal{M}}_{0, k+2}$, of dimension $k - 1$. Every x insertion has a positive ancestor power, since $x(z) \in z\widehat{\mathcal{H}}_+$. Their product restricts to degree at least k on this factor and therefore vanishes. This argument, applied to each cotangent difference, replaces the original ancestors by the ones retaining all x marks. Multilinearity now gives (4.16).

The assertion uses the stable-curve convention for ancestor potentials. Terms which become stable only after adding x marks do not introduce a nonconstant correction: in genus zero their

positive ancestor powers exceed the dimension of the stable-curve space, and in genus one the case without an original distinguished mark contributes only a scalar. This accounts for the scalar allowed in the statement. $\square$

4.4 Assembly of the stable graphs

Consider a fixed stable map contributing to an ancestor invariant of W with stable distinguished curve data. Forget the map and stabilize its distinguished marked curve. Every moving leg disappears, because it has at most two special points. On the graph whose vertices are entire fixed-map factors, prune unmarked rational trees and suppress rational chains with two remaining flags, keeping all distinguished markings. A fixed-map factor which survives is retained with its own stabilization at those flags and markings; its internal stable-map boundary remains part of its moduli space. The pieces removed between surviving vertices are bridges, those ending in a distinguished marking are tails, and all other removed pieces are ends.

This decomposition identifies the restriction of the global ancestor classes. At a surviving vertex they are its ancestors for the remaining distinguished flags and marks, with the end marks forgotten. For a marking on a tail, its global ancestor is the ancestor at the tail's attachment after contraction. These statements follow from the composition of the stabilization maps with gluing of the surviving pointed curves.

Ends therefore supply the background ϵ_F at each vertex. Lemma 4.3 converts its bridge and tail denominators to ancestor denominators and inserts the factors $S_f(a)$. Bridges become exactly $D(\bar{\psi}, \bar{\psi}')$. The direct external markings together with all tails become, by (4.10),

$$R(-z)^*t(z) = R(z)^{-1}t(z), \quad z = \bar{\psi}.$$

Finally (4.16) replaces the background by u_F and adds x_F . In view of (4.12), the complete vertex substitution is consequently

$$t(z) \mapsto R(z)^{-1}t(z) + z(1 - R(z)^{-1}1). \quad (4.17)$$

The upper-triangular quantization formula [9, Proposition 7.3] is now applicable. If

$$D(s, s') = \sum_{i,j \geq 0} D_{ij} s^i (s')^j,$$

its contraction operator is $\hbar/2$ times the second derivatives with coefficient tensors D_{ij} , using the categorical inverse of the twisted pairing. Exponentiating this operator on the product of the fixed ancestor potentials and then making (4.17) is U_R . Indeed its kernel is (4.14); the arguments $-w, -z$ are the signs required by the negative Darboux modes $(-z)^{-i-1}$. This is also [3, Equation (19)] in the present notation.

The localization graph sum has precisely this form. Labeling the flags first and subsequently dividing by permutations gives its factorials and graph automorphism factors. A bridge joining two vertices, or two flags at the same vertex, contributes one factor of $\hbar$ under the same contraction rule; cycles therefore have the required genus power. All pairings and permutations are those in super vector spaces. The tensor gluing proof consequently includes odd cohomology without a change of pairing convention. Components with no distinguished markings can change the overall scalar, which is invertible as a formal exponential. We have proved (4.2).

It remains to check the invertibility asserted in the proposition. The series $R - \text{id}$ has positive filtration, so its inverse and logarithm are defined successively in Novikov degree. The same holds

for the actions obtained by exponentiating the corresponding quadratic operators. At a fixed curve coefficient, genus power, and number of variables, only finitely many filtration-positive factors can contribute. The remaining sums over ancestor indices are finite because an ancestor on a stable n -pointed genus- g curve has total degree at most $3g - 3 + n$; in the fixed descendant identities the corresponding bound is the finite dimension of the fixed-target stable-map space. The inverse upper action has the same properties. Thus all transformations used here and their inverses act in the stated coefficientwise completion. This completes the proof of Proposition 4.1.

5 A grading and a shift in genus zero

We now construct two commuting operators on the equivariant quantum cohomology of W . One is the grading operator. The other translates λ by the loop variable z and is defined by genus-zero invariants of a bundle over $\mathbb{P}^1$. Its fixed-locus expression will connect the ordinary theories of B and X . Its spectrum, computed below using only fiber curves, will provide the branches along which we transport the constraints.

Throughout this section, $\star$ means the small equivariant quantum product of W , at primary background zero. Put

$$c = c_1^T(TW), \quad \kappa = c_1(TB) + c_1(E).$$

The equivariant projective-bundle formula gives

$$c = (r+1)p + \pi^*\kappa + \lambda, \quad \int_{(\beta,l)} c_1(TW) = (r+1)l + \int_{\beta} \kappa. \quad (5.1)$$

Choose a homogeneous Hodge basis of $H^*(B)$ and the resulting polynomial basis $\pi^*b p^j$, $0 \leq j \leq r$, of $H_T^*(W)$ over $\mathbb{C}[\lambda]$. In this basis, μ acts by the first Hodge degree minus $\dim_{\mathbb{C}}(W)/2$ and does not differentiate the coefficient λ .

5.1 The calibrated grading

Define

$$\begin{aligned} G_d &= z\partial_z + \lambda\partial_{\lambda} + \frac{1}{2} + \mu + \frac{c\cup}{z}, \\ G_a &= z\partial_z + \lambda\partial_{\lambda} + \frac{1}{2} + \mu + \frac{c\star}{z}. \end{aligned} \quad (5.2)$$

The subscripts distinguish the constant, or descendant, frame from the quantum, or ancestor, frame. Both operators act on coefficients in λ , whereas the Novikov variables are held fixed.

Lemma 5.1. *With $S_W = S_W(0, z)$ in the convention fixed above,*

$$G_a = S_W G_d S_W^{-1}. \quad (5.3)$$

Proof. Let D_c be the derivation of the Novikov ring defined by

$$D_c(Q^{\beta}y^l) = \left((r+1)l + \int_{\beta} \kappa \right) Q^{\beta}y^l.$$

Homogeneity of the two-point invariants defining S_W says

$$(z\partial_z + \lambda\partial_{\lambda} + D_c)S_W = S_W\mu - \mu S_W. \quad (5.4)$$

This identity uses the first Hodge degree. The evaluation maps are algebraic morphisms, and the virtual class and cotangent-line classes are algebraic. Their push-pull operations therefore have the Hodge bidegrees prescribed by virtual dimension, even when the inserted classes are not algebraic. Concretely, if the input and output basis vectors have first Hodge degrees h_j and h_i , the coefficient of $Q^\beta y^l z^{-k-1}$ in the corresponding entry of S_W has λ -degree

$$h_j + k + 1 - h_i - \left((r+1)l + \int_\beta \kappa \right).$$

Since λ and z both have bidegree $(1, 1)$ for this calculation, this is precisely (5.4). It applies equally to primitive and odd inputs.

The divisor equation gives

$$zD_c S_W = (c\star)S_W - S_W(c\cup). \quad (5.5)$$

The scalar equivariant part of c occurs identically in the two multiplications, so cancels in their difference. Substituting (5.5) into (5.4) gives $G_a S_W = S_W G_d$. These identities first hold in the expansion at $z = \infty$. Coefficientwise rationality of S_W , established by the localization calculation, also gives the identities in its Laurent expansion at $z = 0$. $\square$

5.2 The shift operator and its fixed-locus formula

Here is the genus-zero input from Iritani that we use. For a smooth projective variety Y with a $\mathbb{C}^*$ -action, form its associated bundle $\widehat{Y} \rightarrow \mathbb{P}^1$. Let $\sigma_{\min}$ be the section class of the fixed component with positive normal weights. The operator defined by two-fiber section invariants, with monomials $Q^{\widehat{d}-\sigma_{\min}}$, is an unlocalized equivariant cohomology operator. After reversing the sign of Iritani's loop variable, it translates λ to $\lambda + z$ and satisfies

$$T_a = S_Y T_d S_Y^{-1}, \quad T_d|_F = Q^{\sigma_F - \sigma_{\min}} \prod_{m,x} \frac{\prod_{a=-\infty}^0 (x + m\lambda - az)}{\prod_{a=-\infty}^{-m} (x + m\lambda - az)} e^{z\partial_\lambda}. \quad (5.6)$$

Here x runs through the ordinary Chern roots of the weight- m normal subbundle of F ; each quotient has only finitely many remaining factors. This is the projective specialization of [13, Proposition 2.2, Remark 3.4, Definition 3.9, Remark 3.10, Definition 3.13, and Theorem 3.14]. The source's seminegativity condition is automatic for a complete target, since its global functions are constant. Its calibration M is related to ours by $S_Y(z) = M(-z)^{-1}$.

We describe the associated geometry in this application to fix both the section classes and the signs. For the action that has weight one on the trivial summand of $E \oplus \mathcal{O}$, it is

$$\widehat{W} = \mathbb{P}_{B \times \mathbb{P}^1} (\mathrm{pr}_B^* E \oplus \mathrm{pr}_{\mathbb{P}^1}^* \mathcal{O}(-1)). \quad (5.7)$$

Equivalently it is $W \times (\mathbb{C}^2 \setminus \{0\})/\mathbb{C}^*$, with $s \cdot (w, v_1, v_2) = (s \cdot w, s^{-1}v_1, s^{-1}v_2)$. An additional torus scales v_2 , and its equivariant parameter is Iritani's loop variable before sign reversal. The fibers over $[1 : 0]$ and $[0 : 1]$ carry the original action and the action composed with this additional torus. Identifying their equivariant cohomologies by the induced change of torus variables accounts for the translation in (5.6).

More explicitly, write ζ for that parameter before sign reversal, and let $\rho_0(t, u)w = t \cdot w$ and $\rho_1(t, u)w = (tu) \cdot w$ be the actions on the two fibers. The change of torus variables induces $\Phi_1 : H_{T \times \mathbb{C}^*, \rho_0}^*(W) \rightarrow H_{T \times \mathbb{C}^*, \rho_1}^*(W)$ with

$$\Phi_1(f(\lambda, \zeta)a) = f(\lambda + \zeta, \zeta)\Phi_1(a).$$

If ι_0, ι_∞ denote the fiber inclusions, the section correspondence is defined by

$$(\tilde{T}a, b)_\infty = \sum_{\beta, l} Q^\beta y^l \langle \iota_{0*} a, \iota_{\infty*} b \rangle_{0, 2, \sigma_{\min} + (\beta, l)}^{\widehat{W}, T \times \mathbb{C}^*}. \quad (5.8)$$

Only effective section classes are included. The inputs belong to the equivariant cohomologies of their respective fibers, and the pairing on the left is the one on the second fiber. The operator used here is $T_a = (\Phi_1^{-1} \tilde{T})|_{\zeta = -z}$. Its semilinearity translates λ to $\lambda + z$, as asserted.

In this construction the normal weight at X is positive, so X gives $\sigma_{\min}$. A section obtained from a point of X has tautological degree zero in (5.7); a section obtained from the trivial-summand fixed component B is the line $\mathcal{O}(-1)$ and has tautological degree one. Both have base class zero. Thus $\sigma_B - \sigma_{\min}$ is precisely the fiber-line class of W .

This also checks the curve labels in the shift theorem. The integral projective-bundle splitting identifies $H_2(W, \mathbb{Z})$ by (β, l) . The same splitting on (5.7) identifies a section class by $(\beta, 1, l)$. Subtraction of $\sigma_{\min} = (0, 1, 0)$ leaves $(\beta, 0, l)$. The inclusions of the two fibers preserve β and tautological degree, hence induce the same identification on integral homology. At every splitting of a section stable map the labels therefore add in actual integral homology. In particular the use of (5.6) does not replace classes by their numerical equivalence classes.

Proposition 5.2. *There is a shift operator*

$$T_a = A(z, \lambda, Q, y) e^{z\partial_\lambda}$$

on the small quantum-cohomology module of W , whose matrix A is polynomial in z, λ at every curve coefficient in the chosen equivariant basis. Its calibration is

$$T_a = S_W T_d S_W^{-1}, \quad (5.9)$$

where, writing $v = x + w_F \lambda$ for the equivariant normal roots,

$$T_d|_F = y^{j_F} \prod_x \begin{cases} x + \lambda, & F = X, \\ (x - \lambda - z)^{-1}, & F = B \end{cases} e^{z\partial_\lambda}, \quad j_X = 0, \quad j_B = 1. \quad (5.10)$$

All operators act on the full equivariant cohomology with its super pairing.

Proof. The target W is smooth and projective, hence satisfies the semiprojectivity and weight hypotheses of the cited theorem. Its equivariant formality is also explicit in the basis $\pi^* b p^j$. Formula (5.7) is projective, so its section invariants are defined by proper equivariant pushforward. The two fiber insertions are polynomial equivariant classes, and the equivariant pairing of W is perfect over $\mathbb{C}[\lambda]$. Consequently the operator is polynomial in both equivariant parameters at each curve coefficient; changing the sign of the second parameter and identifying the fibers preserves this property.

The normal weights are 1 at X and -1 at B . In (5.6) these give the factors v and $(v - z)^{-1}$, respectively. The section calculation above gives y^{j_F} . Finally, in the convention for the fundamental solution used here, its pairing formula and symplectic identity give $S_W(z) = M(z)^* = M(-z)^{-1}$. Iritani's identity $MT_a = T_d M$, with his z replaced by $-z$, is therefore exactly (5.9).

The two-marked section correspondence and its localization proof use arbitrary evaluation classes and diagonal contractions. They thus apply on full cohomology with the categorical inverse pairing and its Koszul signs. At primary background zero no assertion about a power-series extension in odd background coordinates is needed. $\square$

Lemma 5.3. *For every fixed actual base class β , the matrix coefficients of A and of $c\star$ are polynomials in y, z, λ (with no z dependence in $c\star$). In particular they are holomorphic functions of y before any localization in λ .*

Proof. The dual of the vector bundle in (5.7) is the sum of the globally generated bundles $\mathrm{pr}_B^* E^*$ and $\mathrm{pr}_{\mathbb{P}^1}^* \mathcal{O}(1)$. Its tautological line bundle is therefore globally generated. Every effective section class has $l \geq 0$, and subtracting $\sigma_{\min}$ does not change l . The same lower bound for W was established by the master-space construction.

Fix basis vectors for an input and a paired output. The section invariant defining the corresponding shift entry has two fiber incidence insertions of fixed cohomological degrees. The projective-bundle formula on $\widehat{W}$ gives

$$\mathrm{vdim}_{\mathbb{C}} \overline{\mathcal{M}}_{0,2}(\widehat{W}, \sigma_{\min} + (\beta, l)) = \dim_{\mathbb{C}} W + 1 + (r+1)l + \int_{\beta} \kappa. \quad (5.11)$$

Thus virtual dimension increases by $r+1$ when l increases by one. Proper equivariant integration is polynomial in the parameters. If virtual dimension exceeds the total degree of the insertions, its result would have negative degree and is zero. Thus only finitely many l occur for this entry and β . The finite polynomial inverse-pairing matrix used to recover operator entries does not change this conclusion. The identical argument with the three primary insertions defining a quantum product proves it for $c\star$. $\square$

We have now obtained operators whose coefficients can be studied both as Novikov series at $y = 0$ and as functions of a nonzero complex variable y . Before making that change of viewpoint, we verify that the shift respects the grading and determine its limiting spectrum.

Lemma 5.4. *The grading and shift commute:*

$$[G_a, T_a] = 0. \quad (5.12)$$

Proof. By the two calibration identities it suffices to work in the descendant frame. Equivariant homogeneity of restriction identifies $\lambda \partial_{\lambda} + \mu$ on the summand for F with $\lambda \partial_{\lambda} + \mu_F - n_F/2$. Also

$$c|_F = c_1(TF) + \sum_x (x + w_F \lambda).$$

Write $T_d|_F = a_F E_z$, where $E_z = e^{z \partial_{\lambda}}$ and a_F is the multiplier in (5.10). The operator $z \partial_z + \lambda \partial_{\lambda}$ commutes with E_z . Combining it with first Hodge degree counts 1 for each factor $x + \lambda$ and -1 for each factor $(x - \lambda - z)^{-1}$. The resulting commutator with a_F is $W_F a_F$, where $W_F = n_F w_F$. The power y^{j_F} is held fixed in this calculation. On the other hand,

$$E_z(c|_F) = (c|_F + W_F z) E_z, \quad [c|_F/z, a_F E_z] = -W_F a_F E_z.$$

The two contributions cancel. Scalar degree-centering terms and the ordinary cup multiplications in this computation introduce no further commutators. $\square$

5.3 The spectrum of the fiber theory

Let $Q^{>0}$ denote the ideal of positive base degree in the Novikov ring. Since an effective nonzero base class has positive ample degree, reduction modulo this ideal retains exactly the maps with constant projection to B . The variable y is retained. We next identify T_a modulo z in this fiber theory.

Lemma 5.5. *On $H_T^*(W)$ with the fiber quantum product,*

$$T_a \bmod (z, Q^{>0}) = (p + \lambda) \star. \quad (5.13)$$

Proof. The product is linear over the classical superalgebra $H^*(B)$: all evaluations of a genus-zero fiber map have the same projection to B , so base classes pull out of the corresponding pushforward with the usual signs. The same reasoning applies to the two-point operator S_W in this reduction. Formula (5.9), whose shift fixes base cohomology, then shows that T_a is $H^*(B)$ -linear.

Put $D_y = y\partial_y$. The divisor equation gives

$$S_W(zD_y - p\cup)S_W^{-1} = zD_y - p\star.$$

The operator on the left before conjugation commutes with T_d . On X , the multiplier has no y factor and $p|_X$ is independent of λ . On B , $p|_B = -\lambda$ and

$$[zD_y, T_d|_B] = zT_d|_B, \quad [\lambda, T_d|_B] = -zT_d|_B.$$

Thus the two terms cancel there as well. Conjugating and reducing modulo z proves that $T_a|_{z=0}$ commutes with $p\star$.

Assign complex cohomological degree one to p, λ, z and degree $r + 1$ to y for the present fiber calculation. Polynomiality in λ implies that the quantum powers of p through p^r are the classical powers: a positive y contribution has degree at least $r + 1$. These powers form a basis over $H^*(B)[\lambda]$, so an $H^*(B)[\lambda]$ -linear operator commuting with $p\star$ is determined by its value on 1.

In this same grading T_a has degree one. To see this directly from the section definition, the normal bundle to a minimal section has one summand $\mathcal{O}(-1)$ and the other summands are trivial. Hence $c_1(T\widehat{W}) \cdot \sigma_{\min} = 1$. Equivalently, (5.11) at $\beta = 0$ has virtual dimension $\dim W + 1 + (r + 1)l$. Each fiber inclusion raises insertion degree by one; comparing the resulting degree with the fiber pairing gives operator degree $1 - (r + 1)l$ in the coefficient of y^l . It follows that $T_a(1)|_{z=0}$ has no positive y coefficient. At $y = 0$ all curve classes are zero, $S_W = 1$, and (5.10) applied to 1 has restrictions $p + \lambda$ on X and zero on B . These are exactly the restrictions of $p + \lambda$ on W . Thus $T_a(1)|_{z=0} = p + \lambda$, which proves (5.13). $\square$

Let $I = H^{>0}(B; \mathbb{C})$. This is a nilpotent ideal, also when $H^*(B)$ has odd classes. Reducing the fiber quantum algebra modulo I gives

$$\mathbb{C}[\lambda, y, p] / (p^r(p + \lambda) - y). \quad (5.14)$$

Indeed the classical projective-bundle relation reduces to $p^r(p + \lambda) = 0$. In degree $r + 1$ the only possible positive curve correction is a constant multiple of y . Its coefficient is the degree-one line invariant in a fiber $\mathbb{P}^r$, hence one. More explicitly, one may pair the relevant structure constant with a top-degree base class; the degree-zero base projection reduces the computation to the fiber. The base directions contribute no obstruction because $H^1(C, \mathcal{O}_C) = 0$ for a genus-zero domain. The coefficient has degree zero, so is independent of λ and may equally be computed at $\lambda = 0$. It is then

$$\langle H, H^r, H^r \rangle_{0,3,\text{line}}^{\mathbb{P}^r} = 1.$$

Here H is the ordinary hyperplane class of the fiber. Indeed two general point conditions determine a unique line, and the remaining marked point is its unique intersection with a general hyperplane. Convexity of projective space makes this the virtual count as well. This gives the coefficient of y in $p \star p^r$, hence the coefficient in (5.14).

Proposition 5.6. *For generic (y, λ) , the distinct eigenvalue branches of $T_a \bmod (z, Q^{>0})$ are*

$$\lambda + h, \quad h^r(h + \lambda) = y. \quad (5.15)$$

Each branch may have multiplicity and a nontrivial nilpotent part. For $\lambda \neq 0$, near $y = 0$ there is one branch tending to zero and a cluster of r branches tending to λ .

Proof. On the quotient by I , (5.13) and (5.14) give the stated eigenvalues. The finite filtration by powers of I is preserved by the operator, and the induced operator on each successive quotient is the same fiber multiplication tensored with I^m/I^{m+1} . Its characteristic polynomial therefore has the same set of roots. Away from the critical values of $h \mapsto h^r(h + \lambda)$ these roots are distinct. At $y = 0$ the roots of that polynomial are $-\lambda$ and 0 , with multiplicities one and r , respectively; translation by λ gives the final assertion. This argument uses nilpotence of positive-degree base cohomology and does not impose semisimplicity on its quantum cohomology. $\square$

The polynomial dependence of the two operators and the finite set of eigenvalue branches are the ingredients needed for the spectral construction in the next section. The calibration identities retain their full curve labels, so the later return to $y = 0$ will still compare the individual descendant theories of the two fixed manifolds.

6 Spectral projections and block gradings

The shift operator T_a distinguishes $r + 1$ spectral blocks at generic fiber parameter, whereas the localization formula distinguishes the two fixed manifolds B and X . We construct operators on the individual spectral blocks and compare their sum near $y = 0$ with the fixed-manifold blocks. The construction must also remove differentiation in the equivariant parameter: only after that removal can the resulting loop operators be quantized over the coefficient ring. All assertions in this section are classical statements about operators.

6.1 Functional calculus for differential series

Let V be the finite-dimensional vector space underlying a chosen equivariant basis of $H_T^*(W)$. On a small open set U in the (y, λ) -plane, write $\mathcal{O}_U$ for holomorphic functions and

$$\mathcal{D}_\lambda(U) = \left\{ \sum_{j=0}^J a_j(y, \lambda) \partial_\lambda^j : J < \infty, a_j \in \mathcal{O}_U \right\}.$$

The multiplication is composition, so $\partial_\lambda a = a \partial_\lambda + \partial_\lambda(a)$. There is no differentiation in y . Denote by Λ_B the base Novikov completion already fixed, and by $\Lambda_{B,+}$ its positive degree ideal. We use the completed algebra

$$\mathcal{A}_U = \left\{ \sum_{\beta, m \geq 0} Q^\beta z^m A_{\beta, m} : A_{\beta, m} \in \text{End}(V) \otimes \mathcal{D}_\lambda(U) \right\}. \quad (6.1)$$

Here and below the notation $\beta \geq 0$ means that β ranges over the effective base labels, including zero. The completion is taken with respect to z and ample base degree. In particular each $A_{\beta, m}$ has finite differential order, without a bound uniform in β, m . Products at a fixed coefficient are

finite by Novikov finiteness. The element z is central in this algebra. We may equally work with germs, and then all holomorphic assertions are coefficientwise. When a loop derivative occurs, we adjoin $z\partial_z$ with

$$[z\partial_z, Q^\beta z^m A_{\beta,m}] = mQ^\beta z^m A_{\beta,m};$$

all expressions involving this extra derivative have finite order in it.

Suppose $T \in \mathcal{A}_U$ has order-zero reduction

$$T^{(0)} = T \bmod (z, \Lambda_{B,+}) \in \text{End}(V) \otimes \mathcal{O}_U.$$

After shrinking U , choose contours in the complex ξ -plane that avoid the spectrum of $T^{(0)}$ and enclose specified groups of its eigenvalues. The contours are held fixed while y and λ vary in U . Set $R_0(\xi) = (\xi - T^{(0)})^{-1}$ and $K = T - T^{(0)}$. The formal resolvent is

$$R_T(\xi) = (\xi - T)^{-1} = \sum_{n \geq 0} (R_0(\xi)K)^n R_0(\xi). \quad (6.2)$$

Each factor K increases the joint z /base filtration. Consequently each coefficient in this expression is a finite sum of differential operators, and is meromorphic in ξ , with poles only at the eigenvalues of $T^{(0)}$.

Lemma 6.1 (Coefficientwise holomorphic calculus). *For T as above and a scalar holomorphic function f on a neighborhood of $\text{Spec}(T^{(0)})$, held fixed independently of the parameters, define*

$$f(T) = \frac{1}{2\pi i} \int_{\Gamma} f(\xi) R_T(\xi) d\xi, \quad (6.3)$$

where Γ encloses the whole spectrum within that neighborhood. The neighborhood may be disconnected. This construction is unital, multiplicative, and compatible with holomorphic composition. In particular, the functions equal to 1 on one spectral group and 0 on the others give mutually orthogonal idempotents whose sum is the identity. All these operators commute with T .

If, in addition, $[T, \lambda] = zT$, then

$$[f(T), \lambda] = z(\xi f'(\xi))(T). \quad (6.4)$$

If a differential operator G commutes with T , acts continuously on the coefficientwise Laurent extension of (6.1), and commutes with the scalar spectral variable ξ , then $[G, f(T)] = 0$.

Proof. Both inverse identities for (6.2) follow from the geometric series. Since ξ and a second spectral variable η are central, the resolvent identity is

$$R_T(\xi)R_T(\eta) = \frac{R_T(\xi) - R_T(\eta)}{\eta - \xi}.$$

Integrating on nested contours and applying the scalar Cauchy formula proves multiplicativity. It also proves $1(T) = \text{id}$, either directly or coefficientwise from (6.2). This gives the projector assertions. Commutation with T follows from the inverse identity.

For completeness, multiplicativity also gives the composition rule. For η outside the spectrum of $f(T^{(0)})$, apply the calculus to $(\eta - f(\xi))^{-1}$. Its product with $\eta - f(T)$ is the identity, so it is the resolvent of $f(T)$. Integrating this identity against $g(\eta)$ and using the scalar Cauchy formula yields $g(f(T)) = (g \circ f)(T)$ whenever the functions are defined on the indicated neighborhoods.

All contour operations here are performed on a fixed coefficient, where there are only finitely many differential compositions.

The commutator with λ follows directly from the inverse rule:

$$\begin{aligned} [R_T(\xi), \lambda] &= R_T(\xi)[T, \lambda]R_T(\xi) \\ &= z(\xi R_T(\xi)^2 - R_T(\xi)) = -z\partial_\xi(\xi R_T(\xi)). \end{aligned}$$

Integration by parts proves (6.4). Finally, $[G, T] = [G, \xi] = 0$ implies $[G, R_T(\xi)] = 0$ by the inverse identity. Integration proves the last assertion. Keeping the contours fixed locally justifies differentiating the coefficient germs under the integral. $\square$

We will compare this calculus in frames related by matrices with negative powers of z . The comparison requires a larger algebra, but does not require convergence in the loop variable. Let $\mathcal{O}$ be a ring of holomorphic parameter germs stable under ∂_λ , and let $\mathcal{M}$ be either the base Novikov monoid or the full monoid of labels (β, l) , $l \geq 0$. Write

$$\mathcal{A}_{\mathcal{M}}^{\text{Laur}}(\mathcal{O}) = \left\{ \sum_{d \in \mathcal{M}} q^d \sum_{m \geq m(d)} z^m A_{d,m} : A_{d,m} \in \text{End}(V) \otimes \mathcal{D}_\lambda(\mathcal{O}), m(d) \in \mathbb{Z} \right\}. \quad (6.5)$$

The notation q^d means Q^β or $Q^\beta y^l$, respectively. The Novikov support condition is imposed as before. The lower z -bound can depend on d . Multiplication is well-defined: a fixed label has finitely many decompositions, and for each decomposition the two Laurent lower bounds leave only finitely many summands at a fixed power of z .

Lemma 6.2 (Change of frame for resolvents). *Let T and T' have coefficientwise resolvents on a common contour, constructed as in Lemma 6.1. Suppose these resolvents belong to the same algebra (6.5). If M and M^{-1} belong to that algebra, are independent of ξ , and $T' = MTM^{-1}$, then*

$$R_{T'}(\xi) = MR_T(\xi)M^{-1}, \quad f(T') = Mf(T)M^{-1}$$

for every function represented by that contour calculus.

Proof. The expression $MR_T(\xi)M^{-1}$ is both a left and a right inverse of $\xi - T'$ in the common algebra. It therefore equals its other inverse $R_{T'}(\xi)$. Integrating the equality coefficientwise gives the second assertion. The Laurent lower bounds ensure that each product used in the integration is finite at a fixed coefficient. $\square$

Two forms of M will occur. The first is a matrix equal to the identity in degree zero whose coefficient at each positive Novikov label is the Laurent expansion at $z = 0$ of a rational function of z . Its Novikov inverse has the same property. The second is a matrix of the form $\exp(N/z)M_+(z)$, where N is a nilpotent cup-product operator and $M_+(z)$ and its inverse are regular at $z = 0$. The exponential and its inverse are then Laurent polynomials in z^{-1} . Thus both forms satisfy the Laurent requirement of Lemma 6.2.

6.2 Block operators without equivariant differentiation

We now apply the calculus to $T = T_a$. By Lemma 5.3, its coefficients are polynomial in y, λ, z at fixed base degree before the shift $\exp(z\partial_\lambda)$ is expanded. In particular it belongs to (6.1). By (5.15), the spectrum of its reduction modulo z and positive base degree consists of

$$a_i = \lambda + h_i, \quad h_i^r(h_i + \lambda) = y. \quad (6.6)$$

Take $\lambda \neq 0$, $y \neq 0$, and exclude the values where these roots coincide. On a small open set in this locus, choose their labels and a logarithm near each a_i . A label refers to a whole generalized eigenspace: no diagonalizability within that space is assumed. Define

$$P_i = \frac{1}{2\pi i} \int_{\gamma_i} R_{T_a}(\xi) d\xi, \quad \mathcal{L}_i = \frac{1}{2\pi i} \int_{\gamma_i} \log \xi R_{T_a}(\xi) d\xi. \quad (6.7)$$

Here γ_i encloses the one spectral value a_i with its full multiplicity. For a group of these values we use the sum of the contours; its logarithm is specified on each component.

Proposition 6.3 (Removal of coefficient differentiation). *The operators in (6.7) satisfy*

$$[P_i, \lambda] = 0, \quad [\mathcal{L}_i, \lambda] = zP_i, \quad [G_a, P_i] = [G_a, \mathcal{L}_i] = 0. \quad (6.8)$$

In particular P_i and $M_i = \mathcal{L}_i - zP_i\partial_\lambda$ are matrix series, without coefficient differentiation. The operator

$$D_i = P_i G_a - \frac{\lambda}{z} \mathcal{L}_i \quad (6.9)$$

differentiates only in z and satisfies $P_i D_i = D_i P_i = D_i$.

Proof. The shift form gives $[T_a, \lambda] = zT_a$. Apply (6.4) to the locally constant branch indicator and to its supported logarithm. These give the first two commutators. The relation $[G_a, T_a] = 0$ from (5.12) gives the last two. The operator G_a has only one negative loop power and first-order parameter derivatives, so its action and the commutators used here are defined in (6.5).

For a finite-order differential operator D in λ , $[D, \lambda] = 0$ implies that D has order zero: the commutator of $a_J \partial_\lambda^J$ with λ has leading term $J a_J \partial_\lambda^{J-1}$. Apply this observation at each coefficient of P_i and $\mathcal{L}_i - zP_i\partial_\lambda$. Writing out G_a now gives

$$D_i = P_i z \partial_z + P_i \left(\frac{1}{2} + \mu\right) + z^{-1} (P_i(c\star) - \lambda M_i). \quad (6.10)$$

There are no remaining λ derivatives, and neither G_a nor the functional calculus introduced derivatives in y or Q . The support assertions follow from $[G_a, P_i] = 0$, $P_i \mathcal{L}_i = \mathcal{L}_i P_i = \mathcal{L}_i$, and $[P_i, \lambda] = 0$. $\square$

The subtraction in (6.9) has accomplished the first goal: it gives loop operators over the coefficient ring. We record their precise bounds before making the comparison at $y = 0$.

Proposition 6.4 (Modes and their coefficient bounds). *Define*

$$A_{-1,i} = z^{-1} P_i, \quad A_{k,i} = z^{-1} (z D_i)^{k+1} \quad (k \geq 0). \quad (6.11)$$

Each $A_{k,i}$ is a series in powers z^m with $m \geq -1$, whose coefficients are matrix differential operators in $z \partial_z$ of order at most $k+1$. Its coefficients contain no derivatives in y, λ or Q . Each fixed base and loop coefficient continues holomorphically along paths in the generic locus of (6.6), with the spectral labels and logarithms continued along the path.

For any scalar b independent of z , replacing D_i by $D_i + bz^{-1} P_i$ gives

$$A_{k,i} \mapsto \sum_{j=0}^{k+1} \binom{k+1}{j} b^{k+1-j} A_{j-1,i}. \quad (6.12)$$

For distinct branches $i \neq j$, one has $(z D_i)(z D_j) = 0$. Thus, if $P = \sum_i P_i$ and $D = \sum_i D_i$ for a group of branches, then the modes formed from P, D are the sums of their individual modes.

Proof. Equation (6.10) says that zD_i has nonnegative powers of z and Euler differential order at most one. Since $z\partial_z$ preserves each loop power, products do not create negative powers. This proves the asserted bounds.

At a fixed base and loop coefficient, the resolvent construction uses finitely many matrix inversions, parameter derivatives, and contour residues. Its only spectral denominators come from the eigenvalues in (6.6). The input coefficients are polynomial, so these operations continue along any path on which the eigenvalues remain distinct and nonzero, with the logarithms continued. This is a statement about individual coefficients; it asserts no convergence of the z or Novikov series.

Since D_i differentiates only z , it commutes with b . Furthermore P_i commutes with z and is a two-sided identity for D_i . The ordinary binomial formula in this supported algebra gives (6.12), including its $j = 0$ term $b^{k+1}z^{-1}P_i$. Finally,

$$(zD_i)(zD_j) = zD_iP_i zP_jD_j = 0 \quad (i \neq j),$$

which proves the assertion about sums of branches. $\square$

In particular, changing a logarithm by $2\pi i n$ changes D_i by $-2\pi i n \lambda z^{-1}P_i$. All the modes for one logarithm therefore span the same collection as those for any other logarithm. Infinitesimal symplecticity will follow from the fixed-manifold calculation and continuation; it is not needed for the constructions above.

6.3 The two fixed-manifold blocks at $y = 0$

Fix a germ of $\lambda \neq 0$. At $y = 0$, one eigenvalue $a_B = \lambda + h_B$ tends to zero, while the other r tend to λ . We call the first the B branch and the second group the X cluster. Choose disjoint small contours about 0 and λ that enclose these groups for y sufficiently small. The projector calculus applies through $y = 0$ on both contours. The logarithm applies through $y = 0$ on the X contour, since that contour bounds a neighborhood not containing zero. Denote the resulting operators by $P_B, P_X, \mathcal{L}_X$.

For this comparison express both frames in the fixed-cohomology coordinates furnished by restriction to $B \sqcup X$. The change from the polynomial equivariant basis depends only on λ and is invertible at $\lambda \neq 0$; it preserves the completed algebras and all assertions about holomorphic dependence on y . Let Π_B, Π_X be the constant projections onto these two summands. In these coordinates write

$$K_F(z, \lambda) = \prod_x \begin{cases} x + \lambda, & F = X, \\ (x - \lambda - z)^{-1}, & F = B, \end{cases} \quad E_z = \exp(z\partial_\lambda),$$

where the product is over the ordinary Chern roots of N_F . Thus (5.10) reads $T_d|_F = y^{j_F} K_F E_z$.

Proposition 6.5 (Comparison at zero fiber degree). *The projectors P_B, P_X and the cluster logarithm $\mathcal{L}_X$ have holomorphic coefficient germs through $y = 0$. Their Taylor expansions at $y = 0$ satisfy*

$$P_F = S_W \Pi_F S_W^{-1} \quad (F = B, X), \quad \mathcal{L}_X = S_W \mathcal{L}_{d,X} S_W^{-1}, \quad (6.13)$$

where $\mathcal{L}_{d,X}$ is the logarithm of $K_X E_z$ on the X summand, extended by zero on B .

Moreover, $T_a P_B$ is divisible by y coefficientwise in z and base degree. Set

$$\tilde{T}_B = y^{-1} T_a P_B. \quad (6.14)$$

It has holomorphic coefficient germs through $y = 0$. Its nonzero spectral block at $y = z = Q^{>0} = 0$ has eigenvalue $(-\lambda)^{-r}$, with possible nilpotent part, and its complementary block is zero. If $\mathcal{K}_B$ is its supported logarithm on the nonzero block, then, on a punctured sector in y , one can choose the B -branch logarithm as

$$\mathcal{L}_B = (\log y)P_B + \mathcal{K}_B. \quad (6.15)$$

The coefficients of $\mathcal{K}_B$ are holomorphic through $y = 0$. Its Taylor expansion, and hence (6.15), obeys the same conjugation by S_W as in (6.13), with the corresponding fixed-block operators in the descendant frame.

Proof. The reduction $T_a^{(0)}$ is a holomorphic matrix through $y = 0$. Its spectrum there consists of the two separated values $0, \lambda$, including all their multiplicities and nilpotent parts. The Neumann construction with the two fixed contours therefore gives the asserted holomorphic extensions of $P_B, P_X, \mathcal{L}_X$.

To compare frames, first expand these coefficient germs in y . Taylor expansion is a homomorphism commuting with ∂_λ ; it sends the ancestor resolvent, whose loop powers are nonnegative, to a Laurent series in the full Novikov labels and preserves both of its inverse identities. Use (6.5) with full labels $d = (\beta, l)$, $q^d = Q^\beta y^l$, and coefficients holomorphic in the chosen λ germ. The rationality assertion in Proposition 4.1 puts the expansion of S_W at $z = 0$ in this algebra: a rational function has a finite-order pole at zero. Since its zero-curve coefficient is the identity, its inverse belongs to the same algebra. The two shift operators, expanded using (5.10), also belong to it. Their resolvents on the two contours do as well. For the descendant resolvent one can use the joint filtration by z and full Novikov degree: its initial B block is zero, and its initial X block is multiplication by $\lambda + p$, whose nilpotent part is the ordinary class $p|_X$.

Equation (5.9) and Lemma 6.2 now identify the actual resolvents in the two frames. In the descendant frame the contour about zero selects exactly the whole B summand, and the contour about λ selects exactly the whole X summand. Indeed the reduced spectra have this property; within each summand the contour function is identically 1 or 0, so Lemma 6.1 gives respectively the identity or zero also for the formal perturbation. Integration proves (6.13), including the logarithm on X .

On the B summand, $T_d \Pi_B = y K_B E_z \Pi_B$. The parameter y is central throughout the algebra. Hence

$$T_a P_B = y S_W K_B E_z \Pi_B S_W^{-1}$$

in the Laurent algebra completed in full Novikov degree. Neither S_W nor its inverse has a negative y power. The constant Taylor coefficient in y of every base and loop coefficient on the left therefore vanishes. Those coefficients were already holomorphic through $y = 0$, so division by y gives holomorphic coefficients and

$$\tilde{T}_B = S_W K_B E_z \Pi_B S_W^{-1} \quad (6.16)$$

after Taylor expansion.

At $y = z = Q^{>0} = 0$, the calibration is the identity and the nonzero block of (6.16) is $\prod_x (x - \lambda)^{-1}$. The ordinary positive-degree classes are nilpotent, so its sole spectral value is $(-\lambda)^{-r} \neq 0$. Choose a contour enclosing this value and avoiding zero, and choose a logarithm there. Lemma 6.1 defines $\mathcal{K}_B$ with holomorphic coefficients through $y = 0$. The projector for this nonzero block is P_B , by (6.16) and Lemma 6.2. The same lemma identifies $\mathcal{K}_B$ with the conjugate of the supported logarithm of $K_B E_z \Pi_B$.

Finally restrict to a punctured sector and choose $\log y$. In the algebra supported on P_B one has $T_a = y \tilde{T}_B$. Rescaling the spectral contour by the central scalar y and choosing $\log(y\xi) = \log y + \log \xi$

shows that its logarithm is exactly (6.15). This is a choice of the original contour logarithm in (6.7). The conjugation assertion follows because multiplication by $\log y$ commutes with the entire algebra. $\square$

We have thus constructed the same block operators in two useful forms. At generic y , their individual coefficients continue with the roots in (6.6). Near $y = 0$, they are conjugates of operators on the two fixed manifolds. The coefficients of the B -branch modes are finite polynomials in $\log y$ with holomorphic coefficients. Indeed $D_B = D_B^{\text{reg}} - \lambda(\log y)z^{-1}P_B$, where $D_B^{\text{reg}} = P_B G_a - (\lambda/z)\mathcal{K}_B$ has holomorphic coefficients; apply (6.12) with $b = -\lambda \log y$. For $A_{k,B}$ the logarithmic degree is therefore at most $k + 1$. The X -cluster modes are holomorphic through zero by Proposition 6.5. These statements require neither convergence in z nor an analytic interpretation of the Novikov series.

7 The ordinary theory on a fixed component

The operators of Section 6 were constructed from the equivariant theory of W . We now identify what their equations mean near $y = 0$. On a fixed component, quantum Riemann–Roch removes the inverse Euler twist. The same transformation turns the logarithm of the shift operator into a translation generator in λ . Its derivative term cancels the derivative term of the equivariant grading, leaving the ordinary grading and a scalar multiple of z^{-1} .

Proposition 7.1. *Fix a germ at $\lambda \neq 0$. Near $Q = y = 0$, use the full Novikov completion with a formal $\log y$ adjoined, allowing polynomial dependence on $\log y$ in each coefficient. Let $F = B$ denote the branch near the eigenvalue zero, or let $F = X$ denote the whole cluster near the eigenvalue λ . Use the projectors and logarithms near $y = 0$ of Proposition 6.5, and let $A_{k,F}$ be their modes. Then, as identities over this coefficient ring,*

$$A_W \text{ satisfies every } A_{k,F} \text{ projectively} \iff Z_F \text{ satisfies every } \ell_{k,F} \text{ projectively,} \quad k \geq -1.$$

The operators on the left are infinitesimal symplectic for the equivariant pairing of W . On the right, Z_F is the ordinary descendant potential with its full cohomology and actual integral curve labels.

We first prove the identity of classical operators. We then explain its quantization; this second step requires separating the rank factor of the Euler class before applying quantum Riemann–Roch.

7.1 Removing the normal bundle from the grading

Fix F and abbreviate $n = n_F$, $w = w_F$, and $j = j_F$. Put

$$a = w\lambda, \quad c_N = c_1(N_F)_{\text{ord}}, \quad W_F = nw.$$

After restriction to the descendant space of F , the grading and shift from Section 5 are

$$G_F = z\partial_z + \lambda\partial_\lambda + \frac{1}{2} + \mu_F - \frac{n}{2} + \frac{R_F + c_N \cup + nw\lambda}{z}, \quad (7.1)$$

$$T_F = y^j \prod_x \begin{cases} x + \lambda, & w = 1, \\ (x - \lambda - z)^{-1}, & w = -1 \end{cases} e^{z\partial_\lambda}. \quad (7.2)$$

Here x runs over the ordinary Chern roots of N_F , and multiplication by a class is understood in the second formula. Symmetric expressions in the roots are interpreted by the splitting principle. All expansions in x are finite after evaluation in the cohomology of F .

Choose a germ of $\log a$ at a nonzero value of λ . Define a multiplier Δ_F by

$$\log \Delta_F = \sum_x \left\{ \frac{1}{2} \log(a+x) + \frac{1}{z} \int_0^x -\log(a+t) dt + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} z^{2m-1} \partial_x^{2m-1} (-\log(a+x)) \right\}. \quad (7.3)$$

The B_{2m} are the Bernoulli numbers. The coefficient of z^{-1} has positive ordinary cohomological degree, hence is nilpotent as a multiplication operator. Both Δ_F and its inverse consequently have a finite lower bound on their powers of z . Their positive tails are interpreted as formal series at $z = 0$.

The multiplier maps the ordinary paired loop space of F to the space with pairing η_F^t . Indeed, all terms in (7.3) except the half logarithm are odd in z , so that

$$\Delta_F(-z) \Delta_F(z) = \prod_x (a+x) = e_T(N_F).$$

The inverse Euler factor in η_F^t therefore cancels this product. This also fixes the direction of the square-root factor in (7.3).

For the scalar part of the shift calculation, put

$$L(v) = v - v \log v, \quad \Theta_F(\lambda) = nL(w\lambda). \quad (7.4)$$

Lemma 7.2. *The following identities hold in the formal Laurent calculus of Lemma 6.2:*

$$\Delta_F^{-1} G_F \Delta_F = \ell_{0,F} + \lambda \partial_\lambda + \frac{nw\lambda}{z}, \quad (7.5)$$

$$\Delta_F^{-1} T_F \Delta_F = y^j \exp \left(\frac{\Theta_F(\lambda) - \Theta_F(\lambda+z)}{z} \right) e^{z\partial_\lambda}. \quad (7.6)$$

Under the same change of space, the logarithm prescribed near $y = 0$ becomes

$$z\partial_\lambda + j \log y - \Theta'_F(\lambda) = z\partial_\lambda + j \log y + nw \log(w\lambda), \quad (7.7)$$

up to a scalar constant determined by the logarithm branch.

Proof. The cohomological part of the commutator with μ_F differentiates a function of a Chern root by $x\partial_x$, because x has Hodge type $(1,1)$. Apply

$$z\partial_z + \lambda\partial_\lambda + x\partial_x$$

to one summand in (7.3). The half logarithm contributes $1/2$. If $I(a, x) = \int_0^x -\log(a+t) dt$, then $(\lambda\partial_\lambda + x\partial_x)I = I - x$; thus its contribution is $-x/z$. Each Bernoulli term has total degree zero. Summing over the roots gives $n/2 - c_N/z$, which cancels the two corresponding terms in (7.1). This proves (7.5).

For the shift, add $L(a)/z$ to the summand indexed by x in (7.3). The result depends only on $v = a + x$ and equals

$$\Phi(v) = \left((z\partial_v)^{-1} - \frac{1}{2} + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} (z\partial_v)^{2m-1} \right) (-\log v),$$

where the antiderivative in the first term is $L(v)/z$. The generating series for the Bernoulli numbers gives

$$\Phi(v+z) - \Phi(v) = -\log v. \quad (7.8)$$

When $w = 1$, this difference cancels the factor v in (7.2). When $w = -1$, use instead $\Phi(v-z) - \Phi(v) = \log(v-z)$, which cancels the factor $(v-z)^{-1}$. The terms $L(a)/z$ that were added account for the exponential in (7.6).

It remains to identify the functional logarithm, rather than just one operator whose exponential has the required form. Set

$$H = z\partial_\lambda - \Theta'_F(\lambda).$$

Solving its evolution equation gives

$$e^{tH} = \exp\left(\frac{\Theta_F(\lambda) - \Theta_F(\lambda + tz)}{z}\right) e^{tz\partial_\lambda}. \quad (7.9)$$

For example, differentiating the right side in t gives H times that side and its value at $t = 0$ is the identity. This is also the exponential defined by the holomorphic functional calculus of Lemma 6.1: coefficientwise differentiation of its contour formula gives the same evolution equation. The constant term of H in z is the scalar $-\Theta'_F(\lambda)$. Locally choose a logarithm which inverts the exponential at that scalar. The composition rule in Lemma 6.1 then identifies $\log(e^H)$ with H .

Conjugation of the resolvents by Δ_F is allowed by Lemma 6.2. For $F = B$, apply this argument to the divided shift $y^{-1}T_F$ and then restore $\log y$; for $F = X$, apply it directly to T_F . These are exactly the logarithm prescriptions of Proposition 6.5. This proves (7.7), with the stated freedom in its scalar constant. $\square$

Subtracting λ/z times (7.7) from (7.5) now removes $\lambda\partial_\lambda$. Thus the operator D_F on its fixed block, after the descendant change of space and then Δ_F , is

$$\ell_{0,F} + \frac{b_F}{z}, \quad b_F = \lambda(nw - j \log y - nw \log(w\lambda)), \quad (7.10)$$

up to a scalar multiple of λ/z from a different logarithm choice. In particular, the remaining operator differentiates none of λ , y , or the Novikov variables.

For a scalar b independent of z , write

$$\ell_{-1,F}^{(b)} = z^{-1}, \quad \ell_{k,F}^{(b)} = z^{-1}(z\ell_{0,F} + b)^{k+1} \quad (k \geq 0).$$

The binomial identity

$$\ell_{k,F}^{(b)} = \sum_{m=-1}^k \binom{k+1}{m+1} b^{k-m} \ell_{m,F} \quad (7.11)$$

is triangular with diagonal entries one. Hence these operators are infinitesimal symplectic, and their simultaneous projective equations are equivalent to those of the ordinary modes. We have proved this identification at the classical level. To obtain the corresponding statement for the potentials, we next implement it by quantum Riemann–Roch.

7.2 Quantizing the comparison

The logarithm in (7.3) contains $-\log(w\lambda)c_N/z$. Although this term defines a perfectly good classical multiplication operator, directly exponentiating its quantization would require interpreting a translation that is not small in λ^{-1} . We avoid that interpretation by first using the normalized characteristic class

$$c_0(N_F) = \prod_x (1 + x/a)^{-1}. \quad (7.12)$$

Let $Z_{F,\text{ntw}}$ and η_F^{ntw} denote its descendant potential and pairing. The subscript distinguishes this theory from the inverse Euler twist used in localization.

Define $\Delta_{F,0}$ by the formula (7.3), replacing $-\log(a+x)$ by $-\log(1+x/a)$ and replacing the half logarithm by $\frac{1}{2}\log(1+x/a)$. It maps the ordinary paired space to $(\mathcal{H}_F, \eta_F^{\text{ntw}})$ and is the identity modulo λ^{-1} .

Lemma 7.3 (Quantum Riemann–Roch in the normalized twist). *With the quantization and dilaton translations of Section 2,*

$$Z_{F,\text{ntw}} = (\text{invertible scalar}) U_{\Delta_{F,0}} Z_F. \quad (7.13)$$

This is an invertible identity formal in λ^{-1} and coefficientwise in the full Novikov variables and $\hbar$.

Proof. We use the quantum Riemann–Roch theorem of Coates and Givental [2, Theorem 1]. For a multiplicative characteristic class whose logarithm on a line with first Chern class x is $s(x)$, its multiplier in the fixed ordinary pairing has exponent

$$\sum_x \left\{ \frac{1}{z} \int_0^x s(t) dt + \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} z^{2m-1} s^{(2m-1)}(x) \right\}.$$

The theorem identifies the twisted pairing with the ordinary pairing by multiplication by $\sqrt{c_0(N_F)}$. Expressing its conclusion as a map to the actual twisted space therefore adds $-s(x)/2$ to the exponent. Taking $s(x) = -\log(1+x/a)$ gives exactly $\Delta_{F,0}$.

This change of pairing also specifies the affine coordinates of the quantized action. In the ordinary-pairing Fock space of the theorem, the twisted potential is expressed using

$$\tilde{q} = \sqrt{c_0(N_F)}(t - z1_F).$$

Multiplication by $c_0(N_F)^{-1/2}$ returns the twisted Darboux coordinate

$$q_{\text{ntw}} = t - z1_F.$$

Thus the paired-space map $U_{\Delta_{F,0}}$ has exactly the dilaton translations stipulated in (7.13).

The theorem applies to the universal-curve complex $R\pi_* \text{ev}^* N_F$ on the ordinary stable-map spaces of the smooth projective variety F . Its characteristic class is invertible and formal in λ^{-1} . The quantized multiplier and its inverse are defined in that filtration, with the dilaton translation in each paired space. The scalar factors in quantum Riemann–Roch are immaterial for projective equations.

The super-space formulation in [2, Sections 2–3] uses the full cohomology of the target. In particular, its marked-point terms are even characteristic-class multiplications and its nodal terms contract the categorical inverse of the pairing. It therefore has exactly the Koszul conventions used here, for arbitrary Hodge types and parity. $\square$

The relation between the two classical multipliers is particularly simple:

$$\Delta_F = a^{n/2} \exp(-\log(a)c_N/z) \Delta_{F,0}. \quad (7.14)$$

If C is an ordinary divisor class, then

$$[\ell_{0,F}, C/z] = [z\partial_z, C/z] + [\mu_F, C/z] = 0, \quad [z, C/z] = 0.$$

Here R_F commutes with C , and the two displayed contributions to the first commutator are $-C/z$ and C/z . Consequently C/z commutes with all the ordinary modes and with their scalar translates (7.11). After the coefficient derivatives have been removed in (7.10), the scalar $a^{n/2}$ also cancels in conjugation. Thus Δ_F and $\Delta_{F,0}$ produce the same conjugated mode matrices. By Lemma 7.3 and the covariance of Lemma 2.2, the equations for those matrices on $Z_{F,\text{ntw}}$ are equivalent to the ordinary projective equations on Z_F .

It remains to restore the rank factor omitted in (7.12). This changes the paired space as well as the invariants, and both changes must be made together.

Lemma 7.4. *Put $m_F = a^{-n}$. The actual inverse Euler twist is obtained from the normalized twist by the substitutions*

$$\hbar \mapsto \hbar/m_F, \quad Q^\beta y^l \mapsto Q^\beta y^l a^{-\int_d c_N}, \quad (7.15)$$

where d is the corresponding actual class on F . Its pairing is $\eta_F^t = m_F \eta_F^{\text{ntw}}$. Under these substitutions the quadratic quantizations of $\Delta_{F,0} \ell_{k,F} \Delta_{F,0}^{-1}$, for every $k \geq -1$, in the two paired spaces agree.

Proof. On the moduli space of connected genus- g maps of class d , Riemann–Roch gives

$$\text{rank}(R\pi_* \text{ev}^* N_F) = n(1-g) + \int_d c_N.$$

The ratio of the inverse Euler class to the normalized class of this virtual bundle is therefore

$$a^{-n(1-g)-\int_d c_N} = m_F^{1-g} a^{-\int_d c_N}.$$

This is precisely the change in the coefficient of $\hbar^{g-1} Q^\beta y^l$ under (7.15); exponentiation gives the claimed identity of total potentials. The pairing follows by the same calculation for degree-zero genus-zero three-point invariants.

The qq part of quantization uses the pairing divided by $\hbar$; the pp part uses its inverse multiplied by $\hbar$. In the actual twist these are

$$\frac{m_F \eta_F^{\text{ntw}}}{\hbar}, \quad \frac{\hbar}{m_F} (\eta_F^{\text{ntw}})^{-1},$$

which are their normalized counterparts evaluated at $\hbar/m_F$. The mixed part is unchanged. The matrices $\Delta_{F,0} \ell_{k,F} \Delta_{F,0}^{-1}$ have no Novikov derivatives or Novikov dependence, so the curve substitution commutes with their action as well. We use these unshifted matrices here and restore the scalar b_F afterwards by (7.11). Every actual curve monomial is multiplied by a nonzero scalar; hence the substitution is injective and reversible coefficientwise, without identifying any curve classes. $\square$

The preceding argument used quantum Riemann–Roch only as a formal identity in λ^{-1} . We also need its consequence as an identity of germs at nonzero λ . To make that passage, first remove the scalar b_F by the invertible binomial change (7.11). Each coefficient of the conjugated matrices $\Delta_{F,0}\ell_{k,F}\Delta_{F,0}^{-1}$ is then rational in λ . These matrices have finite lower bounds on their loop powers. The coefficients of $Z_{F,\text{tw}}$ are likewise rational in λ : the torus acts trivially on F , and the inverse Euler class on each fixed stable-map space is expanded only to its finite cohomological dimension. For fixed genus, class, and number of marks, the descendant indices which can occur are bounded for the same reason.

It follows from the finite-support quantization convention that each nonconstant coefficient of

$$Z_{F,\text{tw}}^{-1} \text{op}_{\eta_F^t}(\Delta_{F,0}\ell_{k,F}\Delta_{F,0}^{-1})Z_{F,\text{tw}}$$

is a finite sum of rational functions of λ . Vanishing of its formal Laurent expansion at infinity is equivalent to vanishing as a rational function, and hence as a germ. At a fixed genus and curve coefficient, (7.15) only shifts finitely many powers of λ , so the same argument applies in both directions. We may now restore b_F and its logarithms using (7.11). In particular, no analytic value of the quantized infinite multiplier is being taken in this passage.

Proof of Proposition 7.1. Let F' be the other fixed component. Up to invertible scalar factors, the comparison of potentials follows the chain

$$\begin{aligned} Z_F &\xrightarrow{U_{\Delta_{F,0}}} Z_{F,\text{ntw}} \xrightarrow{(7.15)} Z_{F,\text{tw}}, \\ Z_{F,\text{tw}} &\xrightarrow{U_{S_f}} \mathcal{A}_{F,\text{tw}}(u_F), \\ \mathcal{A}_{F,\text{tw}}(u_F)\mathcal{A}_{F',\text{tw}}(u_{F'}) &\xrightarrow{U_R} \mathcal{A}_W. \end{aligned}$$

The arrow labeled (7.15) changes parameters and rescales the pairing as in Lemma 7.4. The final row first adjoins the other fixed potential and then applies the upper symplectic transformation.

Lemma 7.2, the binomial identity, quantum Riemann–Roch, and Lemma 7.4 identify projective satisfaction of all ordinary modes on Z_F with projective satisfaction of the corresponding descendant modes on $Z_{F,\text{tw}}$. The preceding rationality argument gives this equivalence over the same germs in λ used by the spectral construction.

Pass next from descendants on F to ancestors at u_F . The ancestor–descendant identity of Section 3.3 and Lemma 2.2 conjugate the mode matrices by

$$S_f = S_{F,\text{tw}}(u_F, z)$$

and transport their projective equations to $\mathcal{A}_{F,\text{tw}}(u_F)$. On the product of the two fixed potentials, an operator supported on F acts only on that factor; its projective equation is consequently equivalent to the equation on the single factor.

Finally apply the upper transformation $R(z)$ of Proposition 4.1. Its two identities

$$S_W = R(z) \bigoplus_F S_f, \quad \mathcal{A}_W = (\text{scalar}) U_R \prod_F \mathcal{A}_{F,\text{tw}}(u_F)$$

show that the resulting classical modes are exactly $A_{k,F}$, by Proposition 6.5, and that their quantum equations are the projective equations on $\mathcal{A}_W$. Each change is invertible, so the implication holds in both directions. The classical changes of paired space are symplectic, proving also the symplectic assertion in the proposition.

All the quantized changes in this last paragraph are defined in the Novikov completion: u_F , $S_f - I$, and $R - I$ have positive Novikov filtration. The fixed-theory series S_f has a finite negative tail at each curve coefficient, and R is upper. Conjugation and quantization therefore preserve the lower loop bounds and the finite support required in Section 2. In particular, the comparison uses the upper quantization supplied by localization and the fixed-theory ancestor changes; it does not require quantizing a new expansion of S_W itself. $\square$

8 Continuation and the ordinary constraints

The fixed-component comparison has converted the hypothesis on B into equations on one spectral branch of the ancestor theory of W . We now continue those equations to every branch. The essential finiteness comes from ancestors: their cotangent powers are bounded by the dimension of a moduli space of curves, independently of the fiber degree. We then add the branches belonging to X and use the same fixed-component comparison in reverse.

8.1 Why the equations are identities of germs

Write the connected ancestor potential at background zero as

$$\log \mathcal{A}_W = \sum_{g \geq 0} \hbar^{g-1} \mathcal{F}_g.$$

All coefficients below are taken in the polynomial equivariant basis of Section 3. In particular we have not localized the invariants themselves in order to define them.

Lemma 8.1 (Polynomial coefficients and ancestor bounds). *Fix a genus g , an actual base class β , and a list of insertions $b_i \bar{\psi}_i^{a_i}$, $1 \leq i \leq n$. Their connected invariant in W , summed over fiber degree with weight y^l , is polynomial in y and λ . It vanishes unless the distinguished curve is stable and*

$$\sum_i a_i \leq 3g - 3 + n. \quad (8.1)$$

The latter bound is independent of β, l, λ .

Proof. By definition, curve-unstable ancestor terms are omitted. For stable data, the product of cotangent classes is pulled back from $\overline{\mathcal{M}}_{g,n}$, whose complex dimension is $3g - 3 + n$. This proves (8.1), before integration and without using equivariant degree.

For a class (β, l) , the virtual dimension of the stable-map space is

$$(1 - g)(\dim_{\mathbb{C}} W - 3) + n + \int_{\beta} \kappa + (r + 1)l, \quad \kappa = c_1(TB) + c_1(E).$$

The insertion degrees are fixed. Proper equivariant pushforward takes values in $\mathbb{C}[\lambda]$; it has no negative cohomological degree. Consequently the integral vanishes when this virtual dimension exceeds the total insertion degree. Since $l \geq 0$, only finitely many l can occur. Each of the remaining integrals is polynomial in λ , which proves the claim. This dimension argument uses total cohomological degree; the first Hodge grading used to define the modes is a separate grading. $\square$

The modes on the B branch are infinitesimal symplectic formally at $y = 0$ by Proposition 7.1. By Proposition 6.5, their entries are finite polynomials in $\log y$ with coefficients holomorphic at

zero. Taylor expansion is injective on this ring, so the symplectic identities hold as germs on a punctured sector. This property continues entrywise along any path on which the modes continue: it is a linear identity between their matrix entries and their adjoints. For a branch and logarithm obtained by such continuation, put

$$\mathcal{E}_{k,i} = \mathcal{A}_W^{-1} \text{op}(A_{k,i}) \mathcal{A}_W. \quad (8.2)$$

This notation means that the differential operator acts on the potential, followed by multiplication by its inverse. For the entire X cluster, the notation $\mathcal{E}_{k,X}$ means the same expression with $A_{k,X} = \sum_{i \in X} A_{k,i}$, using a single logarithm throughout the cluster. Projective satisfaction means that every coefficient of positive insertion degree in (8.2) vanishes.

Lemma 8.2 (Finite coefficient tests). *Fix $k \geq -1$, a power $\hbar^{G-1}$, a base class β , and an insertion monomial of degree N in (8.2). Its coefficient is a finite sum of holomorphic germs on the spectral covering of the generic (y, λ) locus, with the chosen logarithms. It continues along every path in that locus. Near $y = 0$, the coefficient for $\mathcal{E}_{k,B}$ is a finite polynomial in $\log y$ with coefficients holomorphic through zero; the coefficient for the cluster expression $\mathcal{E}_{k,X}$ is holomorphic through zero. In these two cases its Taylor expansion is the corresponding formal Novikov coefficient test.*

Proof. Let $F = \log \mathcal{A}_W$ and use positive loop coordinates q . A normally ordered quadratic operator has the schematic form

$$\frac{1}{2\hbar} C(q, q) + \sum_{\alpha, \gamma} B_{\alpha\gamma} q^\gamma \partial_\alpha + \frac{\hbar}{2} \sum_{\alpha, \gamma} D_{\alpha\gamma} \partial_\alpha \partial_\gamma.$$

The tensors have the graded symmetry appropriate to their indices. Dividing its action on e^F by e^F replaces its derivatives by $\partial_\alpha F$ and by the graded expression

$$\partial_\alpha \partial_\gamma F + (\partial_\alpha F)(\partial_\gamma F).$$

The dilaton translation $q = t - z1$ adds only a fixed constant to one coordinate and does not affect finiteness.

At $\hbar^{G-1}$, the first-derivative term involves $\mathcal{F}_G$. The second-derivative term involves $\mathcal{F}_{G-1}$, and the product of first derivatives involves the finitely many pairs $\mathcal{F}_g, \mathcal{F}_h$ with $g + h = G$. Terms with negative genus are absent. The multiplication term occurs only at $\hbar^{-1}$. For a fixed output monomial, the connected potentials which can occur have at most $N + 2$ distinguished marks. Lemma 8.1 therefore bounds every differentiated descendant index uniformly, independently of the fiber degree.

There is also a finite bound on the relevant matrix entries of $A_{k,i}$. By Proposition 6.4 it has loop powers at least -1 and finite order in $z\partial_z$. In the second-derivative part, both negative-mode indices have just been bounded by the ancestor inequality. In the mixed part, the differentiated index is bounded and the multiplied index is either an index of the prescribed output monomial or the dilaton index. In the multiplication part, passage from a nonnegative mode to a negative one, with lower loop bound -1 , permits only finitely many indices. Thus each of the three parts uses only finitely many loop coefficients of the operator. The z derivatives multiply these entries by polynomials in their mode indices and do not change this conclusion.

At fixed β , Novikov finiteness leaves only finitely many decompositions into the base classes carried by the operator and the one or two connected potentials. The latter coefficients are polynomials in y by Lemma 8.1. The relevant operator entries are holomorphic germs that continue

with the spectral branches, by Proposition 6.4. The pairing and its inverse are fixed rational matrices in λ ; we work at a nonzero λ germ. This proves the finite-sum assertion and continuation.

Finally, Proposition 6.5 gives holomorphic coefficients at zero for the X cluster and a polynomial dependence on $\log y$ for each B -branch mode. All operations in the coefficient test have just been shown to be finite. Taylor expansion therefore commutes with that test and yields exactly the formal identity used in Proposition 7.1. $\square$

The last statement is what allows formal identities to start analytic continuation. A finite sum $\sum_{j=0}^J f_j(y)(\log y)^j$, with f_j holomorphic at zero, whose formal series in y and $\log y$ vanishes is zero on a punctured sector: every Taylor coefficient of every f_j vanishes. Thus we use convergent germs of individual coefficient tests, without requiring convergence of a total potential.

8.2 Transport from one branch to the other fixed component

Fix $\lambda \neq 0$. The branch equation is

$$y = h^r(h + \lambda). \quad (8.3)$$

Its critical values are 0 and $(-r)^r \lambda^{r+1} / (r+1)^{r+1}$. Removing these values from the y -plane gives an unramified covering of degree $r+1$.

Lemma 8.3. *The monodromy of (8.3) acts transitively on its $r+1$ branches.*

Proof. The inverse image of the complement of the critical values is the complex h -plane with finitely many points removed. It is connected and path connected. Given any two points over a chosen regular value, join them by a path in this inverse image. Its projection is a loop at that value, whose lift takes the first point to the second. This is transitivity. $\square$

Proposition 8.4. *If the ordinary descendant potential of B satisfies its ordinary modes projectively, then the same is true for X .*

Proof. By Proposition 7.1, the hypothesis gives all the projective equations for $A_{k,B}$ on $\mathcal{A}_W$ near $y = 0$. Lemma 8.2 turns their formal coefficient identities into identities of germs on a punctured sector. Their infinitesimal symplectic identities hold there as well, by the same fixed-component comparison. Symplecticity is an entrywise linear identity, so it continues together with the operators.

Choose a regular value in this sector. Every coefficient of each projective equation continues along any loop based there. The ancestor coefficients themselves return unchanged, because at fixed base class they are polynomials in y . The only change is in the continued spectral branch and logarithm of its mode operator. By Lemma 8.3, the continued B branch can be any of the $r+1$ branches. The projective equations and infinitesimal symplecticity therefore hold on each branch.

Continuing a logarithm may add $2\pi i$ times an integer. Proposition 6.4 gives an invertible triangular binomial change among all modes when this happens. Consequently the simultaneous equations hold for any chosen local logarithm on each branch.

Return to a punctured neighborhood of $y = 0$ and choose the same logarithm on the whole X cluster. Its projectors are the sum of the individual projectors, and its logarithm is the sum of their supported logarithms. In particular $D_X = \sum_{i \in X} D_i$. These operators contain z derivatives, so it is useful to record why taking their powers still respects this sum. Two-sided support and commutation of each projector with z give

$$(zD_i)(zD_j) = zD_i P_i z P_j D_j = 0 \quad (i \neq j).$$

It follows that $A_{k,X} = \sum_{i \in X} A_{k,i}$ for every $k \geq -1$. Quantization is linear, and a sum of quantities independent of the insertion variables is again independent of them. Hence $\mathcal{A}_W$ satisfies all cluster modes projectively.

Proposition 6.5 extends the cluster operators holomorphically through $y = 0$. Taking Taylor coefficients in the equations is legitimate by Lemma 8.2. We may therefore apply Proposition 7.1 in the reverse direction, with $F = X$. Its conclusion concerns precisely the ordinary descendant potential Z_X , with the full curve labels. $\square$

8.3 The prescribed scalar normalization

The use of projective equations has absorbed the scalar ambiguity of quantization. For the ordinary Virasoro operators that ambiguity is removed by two commutators. We give the scalar calculation because the constant in L_0 is part of the assertion.

Lemma 8.5. *For every smooth projective complex variety Y , with the operators of Section 2, one has*

$$[L_{-1}^Y, L_1^Y] = -2L_0^Y, \quad [L_0^Y, L_k^Y] = -kL_k^Y \quad (k \geq -1, k \neq 0). \quad (8.4)$$

Proof. The identities $[\mu_Y, R_Y] = R_Y$ and $[\ell_{0,Y}, z] = z$ give the classical commutators

$$[\ell_{-1,Y}, \ell_{1,Y}] = 2\ell_{0,Y}, \quad [\ell_{0,Y}, \ell_{k,Y}] = k\ell_{k,Y}.$$

Normal ordering in our Hamiltonian sign convention reverses the commutator sign, with a scalar from contractions between the two opposite off-diagonal blocks of the polarization; see [9, Section 2]. We compute the two possible scalars in the full super space.

For the first commutator, multiplication by z^{-1} crosses from the nonnegative to the negative polarization only at mode zero. The nonnegative output of $\ell_{1,Y}$ on the mode $(-z)^{-1}a$ is

$$(1/4 - \mu_Y^2)a$$

in mode zero. Indeed the terms involving R_Y still have negative powers at this output. Contracting the quadratic mode-zero terms in the two orders gives the normal-order constant

$$-\frac{1}{2} \text{str}(1/4 - \mu_Y^2) = -\frac{\chi(Y)}{8} + \frac{1}{2} \text{str}(\mu_Y^2) = -2C_Y.$$

The parity sign can also be checked on Darboux summands. For an even coordinate,

$$[q^2/(2\hbar), \hbar m \partial_q^2/2] = -mq \partial_q - m/2.$$

For an odd paired plane with coordinates θ, ψ ,

$$[\theta\psi/\hbar, -\hbar m \partial_\psi \partial_\theta] = m(1 - \theta \partial_\theta - \psi \partial_\psi).$$

The scalar is $-m/2$ on an even line and m on an odd paired plane, precisely $-\text{str}(M)/2$ for $M = 1/4 - \mu_Y^2$. Together with the classical commutator this is $[L_{-1}^Y, L_1^Y] = -2(\text{op}(\ell_{0,Y}) + C_Y)$.

For the second commutator with $k > 0$, the only crossing of $\ell_{0,Y}$ is its R_Y/z part at mode zero. The opposite crossing of $\ell_{k,Y}$ uses a total of $k - 1$ factors of R_Y : expanding $z^{-1}(z\ell_{0,Y})^{k+1}$, a term that raises loop power by one must have exactly this many zero-power R_Y factors. Its contraction with the first crossing therefore raises first Hodge degree by k . Such an endomorphism has zero trace and zero supertrace. For $k = -1$ both crossings have the same direction, so there is no contraction. Thus the second commutator has no scalar correction. The constant C_Y commutes with every operator, and translating $q = t - z1_Y$ preserves all commutators. This proves (8.4). $\square$

Corollary 8.6. *Projective satisfaction of all the ordinary modes by Z_Y implies $\mathcal{V}(Y)$ with exactly the constant specified in L_0^Y .*

Proof. Projective satisfaction says $L_k^Y Z_Y = c_k Z_Y$, where c_k is independent of all insertion variables. The operators differentiate neither the Novikov parameters nor $\hbar$, so they commute with every c_j . Their commutators consequently annihilate Z_Y . The first identity in (8.4) gives $L_0^Y Z_Y = 0$. The second gives every remaining equation in the required range. $\square$

Proof of Theorem 1.1. The hypothesis $\mathcal{V}(B)$ gives projective satisfaction on B . By Proposition 8.4, it passes to X . Corollary 8.6 gives the prescribed ordinary Virasoro equations.

All constructions used the full cohomology and the categorical inverse pairings. Curve splittings, the shift correspondence, and the final Taylor expansion preserved the actual integral homology labels. Thus the conclusion holds for every genus and curve class, with arbitrary ordinary descendant insertions, as asserted. $\square$

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